

	Occidental Oil & Gas	Approved by: Elliott Heide <i>VP HSE</i>
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OOG 60.400.301 **HSE Risk Assessment & Response Procedure**

Compliance with this procedure is required by the effective date but can be extended with concurrence of OOG Corporate HSE Risk Management. For such cases such as newly acquired facilities, Business Units must submit a written action plan (i.e., road map) to achieve compliance by the agreed date. New requirements defined in this procedure are not retroactive to risk assessment activities conducted prior to the effective date.

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1. PURPOSE

The purpose of this procedure is to provide a means to consistently and systematically identify and assess HSE Risks associated with Occidental Oil and Gas (OOG) business activities; to increase awareness of and manage HSE Risks; and to make HSE Risk Management decisions at the appropriate management levels.

This document provides important information on establishing and maintaining the OOG HSE risk inventory; and developing a framework for the consistent application of sound management of HSE Risks in accordance with the requirements of the OOG HES Risk Management Standard (60.400.300) and Health, Environment and Safety Management System ("HES MS").

2. APPLICABILITY

This procedure is applicable to OOG Business Units (BUs) over which OOG has operational control.

3. REQUIREMENTS

3.1 General Requirements

The OOG HES & SR Policy and HES MS require HSE Risks to be identified, evaluated, reported, and appropriately mitigated. Credible Event Scenarios must be established, and their resulting Consequences and Probabilities must be entered into the OOG HSE Risk Matrix in Appendix A to determine Risk Levels.

The Risk Response specified for each Risk Level must be in accordance with the OOG HSE Risk Response Table - Appendix B.

Each Business Unit must develop and implement a Written Program for HSE Risk Assessment and Response that complies with this document.

The analysis, communications and data used to identify and evaluate HSE Risks, and the resulting Consequence Levels, Probability Levels and Risk Levels constitute "Confidential Company Information" as defined in OPC Policy 10:20:80 ("Confidential Company Information") and are also subject to applicable legal privileges.

Figure 1 - HSE Risk Assessment and Reporting Process, following, shows the overall process for managing risk and the following sections details the requirements for this process.

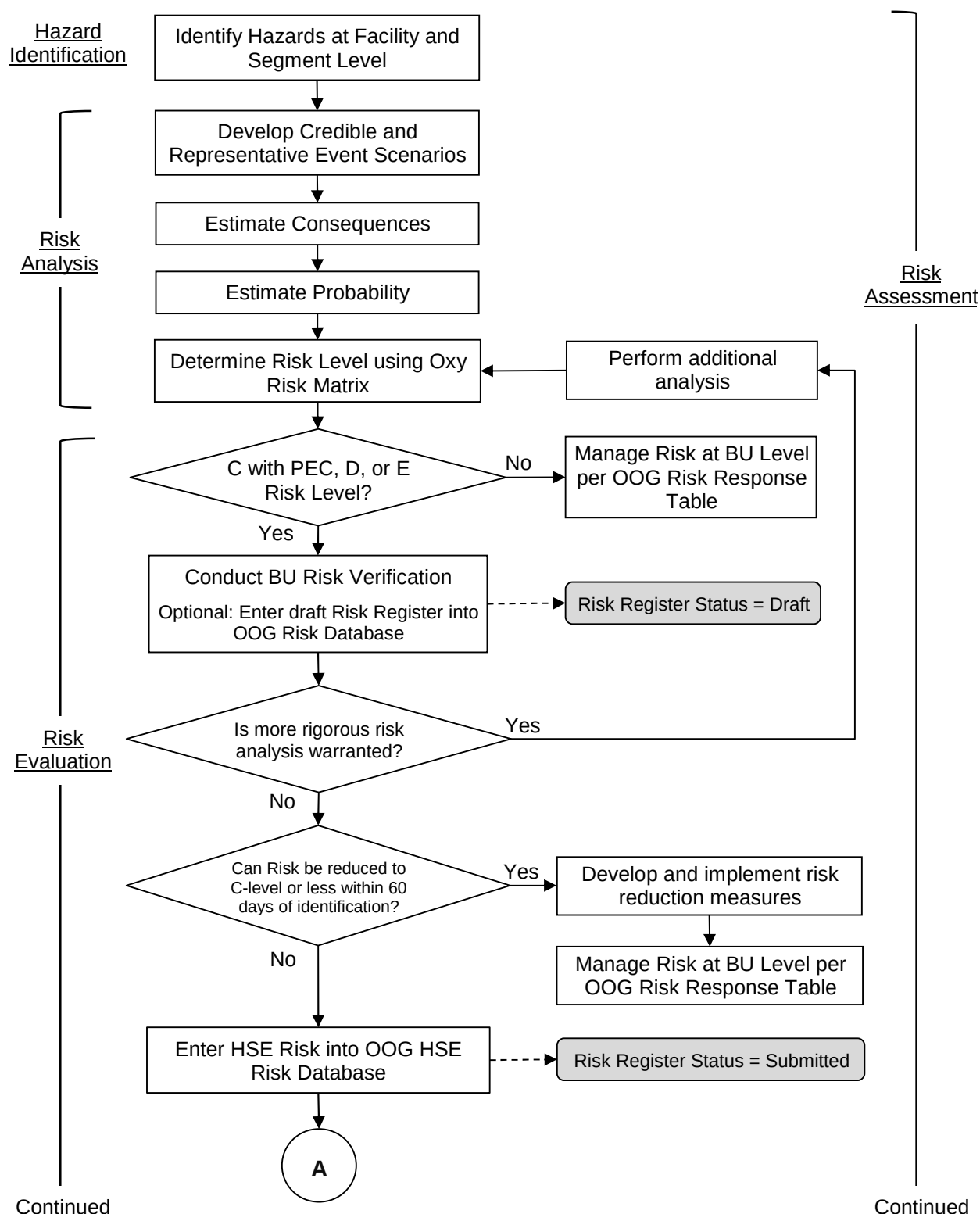
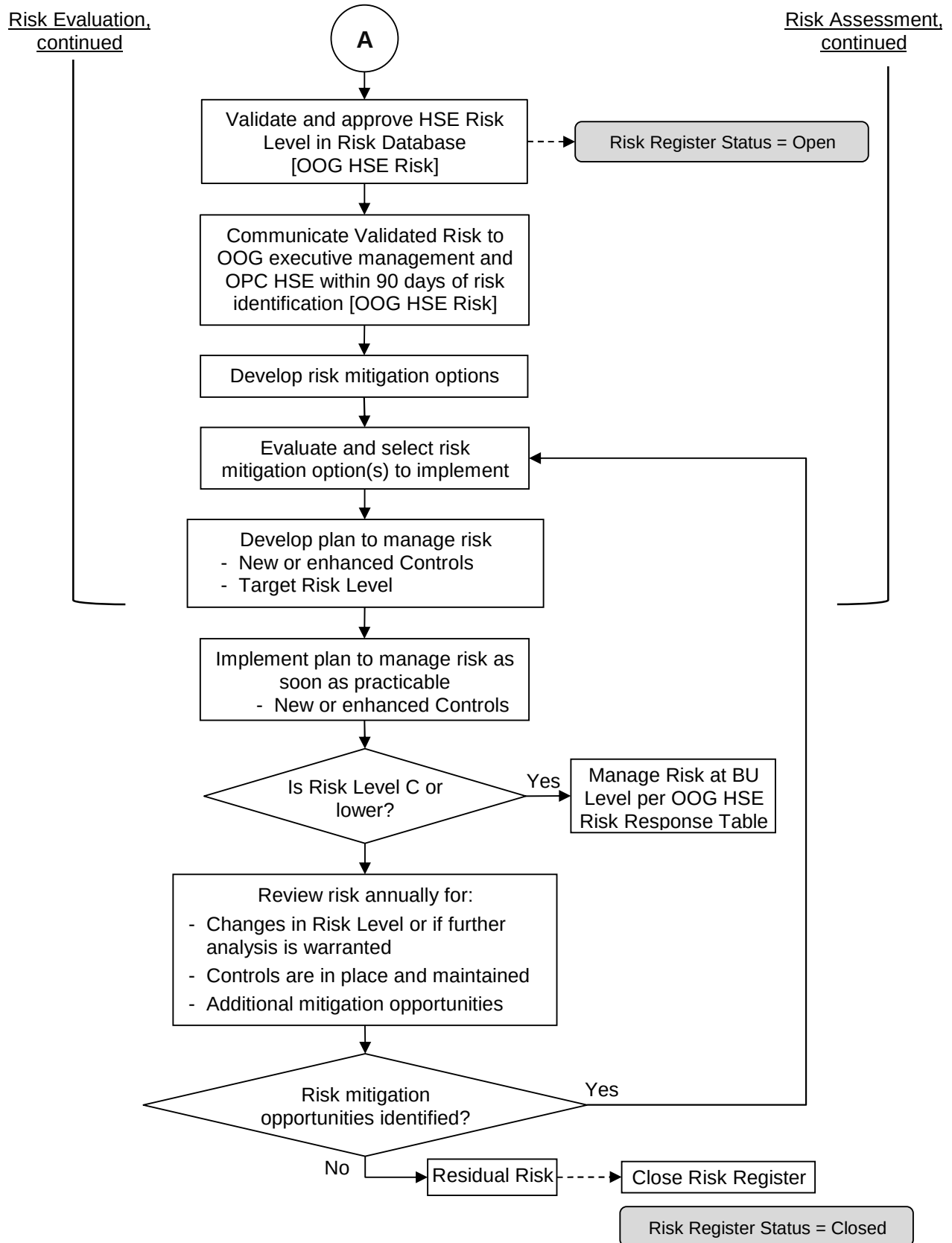


Figure 1 – HSE Risk Assessment and Response Process



3.2 Hazard Identification and Risk Analysis

3.2.1 Identify Hazards

3.2.1.1 Business Units must promote the identification of Hazards at all levels of their organization.

3.2.1.2 Methods to be utilized and sources to be consulted for identification of Hazards must include, but not be limited to, the following:

- a) Multi-disciplinary team hazard reviews such as Process Hazard Analysis (PHA);
- b) Management of Change reviews;
- c) Pre-Start Up Safety Reviews;
- d) Job Observations and/or Behavioral Based Safety Observations
- e) HSE Inspections;
- f) Equipment Integrity Inspections and Performance Testing;
- g) Incident Investigations;
- h) Lessons Learned communications
- i) Internal and external Safety Alerts;
- j) HSE Audits (refer to OOG HES Audit Standard - 60.400.900); and,
- k) Insurance Engineering Surveys

3.2.1.3 Hazard identification must consider, as applicable:

Inherent properties of raw materials, intermediate and final products, and wastes;

Design aspects of processes, equipment, and structures, including facility siting, layout and spacing;

Operations and activities, including construction, operating and maintenance practices, transportation of materials, equipment and personnel, exploration, drilling, well servicing and workover activities, simultaneous operations, off-site activities by third parties, modifications and expansion of facilities, mergers, acquisitions, and divestments;

Extreme operating conditions, location, weather, and other conditions that may give rise to Hazards to people, property, revenue and/or the environment.

Physical and cybersecurity-related aspects of facilities

3.2.2 Develop Credible Event Scenarios

An Event Scenario consists of credible initiating/escalating event(s) that potentially lead to HSE or financial impact consequences. Each Event Scenario must be assessed for each credible Consequence indicated in Appendix A.

Where there is no credible potential for exposure within a Consequence category, there is no credible Event Scenario. See OOG HES Risk Management Guideline

60.400.301.03, “Developing Event Scenarios Guideline.” The HSE Risk Management Program does not address Event Scenarios associated with the use, handling or transportation of products by customers or third parties (other than Contractors).

A Process Event Scenario typically consists of an initiating event (e.g., inadvertent valve closure resulting in overpressure) and a subsequent series of events (e.g., the control system fails to respond and a pressure vessel lacks adequate pressure relief, resulting in loss of containment). After a release scenario is developed, additional series of events or situations are identified that could result in each of the consequences. (Refer to 60.400.301.03 “Development of Event Scenarios Guideline” for guidance.)

3.2.3 Estimate Consequence Levels

Each credible Consequence category of an Event Scenario must be assigned a Consequence Level in accordance with Appendix A. Assignment of Consequence Levels must not take into account Mitigation measures/elements; Mitigation measures/elements must only be taken into account when assigning Probability Levels - see Section 3.1.4 below.

However, if the Event Scenario is credible and not “worst case” (e.g., if people are seldom in the immediate area impacted by a release – for example, a well in an isolated area visited by an operator once per week) the risk evaluator should not automatically assume an injury Consequence. Refer to OOG Guideline 60.400.301.03 for further guidance on credible Event Scenarios.

When a PHA team determines that a combined Property Damage and Loss of Revenue (PD/LOR) Consequence is potentially a Level 3 or higher, the team must estimate the PD and production loss (based on the length and extent of production impairment) and forward their estimates to the BU’s lead risk engineer to assign the final Consequence Level. If the risk engineer does not have the data needed to make a reasonable estimate, he should seek input from the BU engineering department and, if necessary, revise the PD and production loss estimates. The risk engineer will use the final production loss estimates and data from the BU finance manager (typically the Authorization For Expenditure (AFE) long-term assumed product sales pricing estimate) to calculate LOR.

Some Event Scenarios have much higher Consequences than the threshold value for a Level 5 Consequence, an important consideration when making Risk Mitigation decisions. These very high Consequences are referred to as Potential Elevated Consequences. Each event scenario that is evaluated as having a Level 5 Consequence must also be evaluated for Potential Elevated Consequences and assigned one or more PEC Codes as shown in the following table.

PEC 1	At this Event Scenario's assigned Probability, the potential consequence exceeds 14 Permanent Incapacitating Injuries.
PEC 2	At this Event Scenario's assigned Probability, the potential consequence exceeds \$1 billion in combined PD/LOR.
PEC N/A	PEC does not apply to this Event Scenario.

Specialized Field Risk Management (SFRM) risks with Multiple and/or Sensitive Receptors will also be designated as having PECs.

3.2.4 Estimate Probability Levels

Probability Levels represent the presence of Mitigation measures/elements that influence the potential of an Event Scenario occurring.

Typically, in a process facility there is a Probability that containment will be lost, and a subsequent lower Probability of each Consequence occurring (e.g., for an explosion scenario that impacts people, the release must occur, the release must be ignited and people must occupy the overpressure area).

Each Event Scenario must be assigned a Probability Level for each credible Consequence category in accordance with Appendix A using the appropriate methodology below:

Process Risk Facilities	Critical Mitigation Elements Analysis (CMEA) in accordance with Attachment 2
Operational Risk Facilities	Critical Mitigation Elements Analysis (CMEA) in accordance with Attachment 2 unless the CMEA is not appropriate. Drilling and Well Servicing Risks must use Critical Mitigation Systems (CMS) methodology in accordance with Attachment 5. Risk Mitigation Measures (RMM), if approved by OOG Risk, may be used in non-process risk analyses.
Data derived from Quantitative Analysis	If data are utilized from prior internal or external quantitative risk analyses, then the data must be directly applicable and relevant to the Event Scenario and reflect reasonable assumptions. Refer to 60.400.301.03 "Development of Event Scenarios Guideline" for guidance. A quantitative analysis should only be used when: a) a formal qualitative analysis (e.g., CMEA) has been performed by qualified personnel, or b) a CMEA would not be meaningful, or c) quantitative analysis is required to identify or evaluate Mitigation options. NOTE: A QRA must be performed when required by regulation but will not negate the need for CMEA.

3.2.5 Determine HSE Risk Levels

Using the Consequence level and Probability level assigned from the preceding steps, determine the Risk Level (A – E) using the OOG Risk Matrix (Appendix A) for each applicable potential Consequence category. For reporting purposes, the Risk Level of an HES Risk can be considered the highest Risk Level of all consequences analyzed for an Event Scenario.

3.3 Operational Risks

HSE Risks are managed by HSE programs and good practices (e.g., design standards, Operations & Maintenance procedures, Maintenance and Inspection programs, training programs, and emergency response plans) consistent with a Facility's HSE Risks.

Since not all HSE Risks can be anticipated when programs are developed, and HSE Risks can arise from site-specific situations, the Business Unit Manager must take the following additional measures to manage Operational Risks:

- Remind Managers, Supervisors, Employees and Contractors frequently about their responsibility to identify HSE Risks for which practicable Mitigation measures have not been implemented;
- Assign potential and identified Hazards to appropriate persons to evaluate, identify Mitigation options and implement appropriate Mitigation measures;
- Ensure that the OOG HSE Risk Matrix and Risk Response Table (Appendix A and Appendix B, respectively) are used to determine Risk Levels and the appropriate response;
- Evaluate drilling and well servicing risks using the methodology in Appendix E;
- Ensure that an appropriate MOC/PSSR program is maintained, and,
- Reliable means of calculating Probabilities of Operational Risks as described in Section 3.1.4 above are not always available. The following OOG HSE Risks must be assigned default Risk Levels if representative data are not available to estimate their probability:

- a) Security - If a Probability level cannot be established for a security Event Scenario, a default D Risk Level (Consequence level 5, Probability level 2) must be assigned for scenarios that are expected to result in more than three Permanent Incapacitating Injuries outside the Facility, or a lower default level for consequences in which more than three Permanent Incapacitating Injuries are not likely outside the Facility.

If representative data are available to calculate the Probability of a security Event Scenario, the Business Unit must, with the concurrence of OOG HSE Risk Management, determine that Event Scenario's Risk Level using the standard approach instead of applying the default Risk Level.

- b) Aviation Transportation – Due to the significant Consequences associated with transporting personnel via chartered or private aviation transportation, a default D Risk Level (Consequence level 5, Probability level 3) must be assigned to Event Scenarios in which charters carry more than three (3) passengers. A PEC 1 code must be used if the aircraft seats more than 14 passengers and crew, regardless of average occupancy.

Note: Air travel via established international commercial carriers is not evaluated by Business Units since it is a risk that is ranked by OPC as a Level D Risk on a Generic Risk Register for all of Oxy.

If representative data are available to calculate the Probability of an aviation transportation Event Scenario, the Business Unit may, with the concurrence of OOG HSE Risk Management, determine that Event Scenario's Risk Level using the standard approach instead of applying the default Risk Level.

3.4 HSE Risk Evaluation and Response

Risk Levels must be utilized to determine the required Risk Response to HSE Risks in accordance with Appendix A. When a C w/PEC, D or E level risk is identified, immediate action must be taken to ensure that all practical measures are taken to confirm existing mitigations are robust (e.g., operating parameters, AI/MI routines, and alarm / shutdown levels are verified to be appropriate and up to date; interim operating options that would avoid or mitigate the risk are evaluated, etc.).

Risk Responses must include the following actions, as applicable:

- a) Evaluation and Verification;
- b) OOG Risk Validation;
- c) Documentation; and,
- d) Action Tracking.

3.4.1 Evaluation and Verification

If a C, D, or E Level Risk is identified, a qualified risk evaluator must determine if the Risk Level and the PEC codes have been properly assigned and if more analysis is needed to verify the Risk Level and develop Mitigation options.

The Business Unit HSE Risk Management Coordinator or designated qualified risk evaluator must provide BU verification of the Risk Level by:

- Reviewing Consequence and Probability data, including effectiveness and applicability of existing and/or planned Critical Mitigation Elements (CMEs), Risk Mitigation Measures (RMMs) and Key Mitigation Elements (KMEs) essential to the management of the HSE Risk, to ensure proper assignment of Risk Level;
- Ensuring assignment of achievable target dates for implementation of planned CMEs, RMMs and KMEs; and,
- Obtaining Business Unit Management approval of the risk Mitigation strategy and plans.

3.4.2 OOG Risk Validation

OOG Risk validation is a “quality check” performed by OOG HSE Risk Management for all C w/PEC, D and E Level HSE Risks to ensure that appropriate risk data have been provided for making business decisions, identifying and maintaining CMEs, RMMs and KMEs, and identifying additional opportunities for improvement.

The OOG HSE Risk Management validation process must include the following:

- Review of HSE Risk data for completeness and accuracy and when necessary, procure additional information to support and/or clarify HSE Risk data;
- Communication of comments or proposed revisions to the Business Unit HSE Risk Management Coordinator for concurrence.

3.4.3 Documentation

3.4.3.1 OOG HSE Risk Register

Risk registers serve as the mechanism for documenting risk details and mitigation plans / status and for reporting risks to OOG HSE Risk

Management. An OOG HSE Risk Register must be completed for each Event Scenario analyzed as having a Risk Level of C with PEC, D, or E. Risk Registers are generated within the HSE Risk Management Database, assessable from the OOG HSE Risk Community of Practice Team-site for authorized users under Risk Evaluation → Risk Registers, or at <http://hesriskregisters.oxy.com>. Instructions are accessed using the Help button..

Risk Register types must be chosen as follows:

Process Risk Register: Utilized when CME methodology is applied. Process Risk Registers must be designed as a Specific, Master, Subordinate or Generic Risk.

General Risk Register: Utilized when CME methodology cannot be applied. General Risk Registers will be designed as Specific or Generic.

Risk Register Types:

- **Specific:** The designation of a Risk Register when Master, Subordinate, or Generic does not apply.
- **Master:** When there are multiple Loss of Containment Event Scenarios in the same process block or platform, the Risk Register with the highest Risk Level must be designated as the Master.
- **Subordinate:** The Risk Registers within the process block or platform whose risks are represented by the Master Risk Register must be designated as a Subordinate risk register.
- **Generic:** Risk Registers representing substantially similar risks (e.g., third-party excavation of pipelines, risks within the same SFRM Tier Program, etc.) with the same general Mitigation and management plan must be designated as Generic Risk Registers. Generic risk registers may be appropriate at each organizational levels (OPC, OOG, and BU).

3.4.3.2 Risk Inventory

OOG HSE must maintain an inventory of segment-level and Business Unit-level Validated D and E Level HSE Risks and C Level HSE Risks with a PEC Code. Business Units must also maintain an up-to-date inventory of C, D and E Level HSE Risks documented in an appropriate manner (typically Risk Registers for D, E and C w/PEC code, and a spreadsheet for C Level Risks) and make the inventory available for OOG HSE Audits. Documentation must include the following:

- a) Event Scenarios;
- b) Existing CMEs, RMMs and KMEs;
- c) Planned CMEs, RMMs and KMEs; and,
- d) Consequence, Probability and Risk Levels.

3.4.4 Action Tracking

Tracking of planned CMEs, RMMs and KMEs is essential to the success of maintaining up-to-date risk inventories. Mitigation plans, target dates, and cost estimates must be entered in the Planned CMEs/RMMs tab of the Risk Register. Business Units must establish a system to track implementation of planned CMEs, RMMs, and KMEs. Requirements for entry into the OOG approved system must apply for actions/recommendations related to recommendations to address C with PEC, D, and E level risks. KMEs that support CMEs may be grouped, documented, and referenced as a group to improve efficiency. Documentation must include actions taken and dates of completion, and must be made available for OOG HSE Audits.

3.5 ***Risk Reporting and Re-Evaluation***

Business Units must submit a new or updated OOG HSE Risk Register for newly identified and/or changed D and E Level HSE Risks and C Level HSE Risks with PEC Codes to the OOG Director, HSE Risk Management via the HSE Risk Register Database. The HSE Risk Register must be entered into the HSE Risk Register Database within 60 days of identification or a change in position on the OOG HSE Risk Matrix.

The OOG Director, HSE Risk Management must submit a new or updated OOG HSE Risk Register for newly identified and/or changed existing D and E Level HSE Risks and C Level HSE Risks with PEC Codes, to OPC HSE and OOG Executive Management within 90 days of identification or a change in position on the OOG HSE Risk Matrix. For risks identified in existing facility PHAs, the 60-day reporting time will start when the PHA team completes their group meetings. For risks identified in MOCs or PHAs related to facilities that are not in service, the reporting time will be initiated when the facility is placed in service. In all cases, OOG Risk should be consulted when there are atypical situations that justify extending the reporting deadline.

Business Unit must re-evaluate all D and E Level HSE Risks and C Level HSE Risks with PEC Codes at least annually to ensure that existing Controls are effective to support the assigned Risk Level and to determine if additional Mitigation measures are available. Business Units must also ensure that the HSE Risk Registers are kept current. Factors that could influence the Risk Level include new Mitigation measures, changes within the MOC program and associated projects, and Asset Integrity program activities. Possible changes to Risk Level from fluctuations in revenue or facility / system value should be based upon multi-year historical and projected valuations rather than single-year values.

The Business Unit must prepare an annual inventory report on the status of its HSE Risks and HSE Risk Management programs and present the annual risk report to OPC Risk Management in draft by December 31 of the current calendar year and final by January 31 of the next calendar year. The annual report must include the following information (see Business Unit Annual Risk Inventory Report Template – *Attachment 6*):

- a) A cover sheet signed by the General Manager acknowledging agreement with the contents;
- b) A short Executive Summary which summarizes the existing C w/PEC, D and E level risks, major mitigation projects to be undertaken, factors that allow/limit reduction of the BU risk profile; the anticipated BU risk profile in next year's annual risk inventory and progress toward completing all process PHAs using CEMA;
- c) Narrative description of processes associated with each HSE Risks (to allow better understanding of the context of the risk);

- d) Any Regulatory Risk Programs governing the facility, process or operations, including a summary of the basis for applicability, the scenarios addressed (e.g., RMP, Worst Case and Alternate Scenarios), and recent or planned changes in how such programs are managed at the Business Unit;
- e) A qualitative summary of infrequent but recurring conditions or circumstances in which PEC could arise, such as typical Facility activities, meteorological conditions, or community events near a Facility. NOTE: When a QRA's probability for these Consequence Levels are $\leq 1 \times 10^{-8}$, a PEC code is not required.
- f) Overall description of HSE Risk Management activities and accomplishments since the last annual report;
- g) Current Oxy Risk Detail (one-page) reports for C w/PEC, D, and E level risks that are not Residual Risks (see Oxy Risk Detail template at the end of Appendix G).

By January 31, the OPC HSE risk function will review the Business Unit annual risk reports and compile them into a consolidated report or reports for presentation, to and acknowledgement by, the senior OOG executive responsible for the Business Units.

Business Unit management must determine how and when elevated HSE Risks and Risk Mitigation measures are to be communicated to affected employees, contractors, customers and neighboring communities.

4. RESPONSIBILITIES

Responsibilities specific to this document are outlined below. General responsibilities for managing HSE Risks are set out in the HES Management System.

4.1 OOG Director, HSE Risk Management

The OOG Director, HSE Risk Management has responsibility for the development of the OOG HSE Risk Management Annual Report and for validating and submitting to OPC HSE the OOG HSE Risk Registers for new and updated C w/PEC, D and E Level HSE Risks. The Director will keep the OOG VP HSE and OOG Executive Management apprised of the OOG Risk Inventory.

4.2 HSE Risk Management Coordinators

The Risk Management Coordinators (RMC) have responsibility for -

- a) Ensuring that the BU HSE risk management program conforms with OOG HSE risk management procedures;
- b) Keeping BU management apprised of the HSE risk management program's status;
- c) Ensuring that newly identified C with PEC, D, and E-level risks are verified and reported per OOG reporting requirements;
- d) Maintaining the BU HSE Risk Registers current in the HSE Risk Register database;
- e) Compiling and reporting an Annual Business Unit Risk Inventory Report (Appendix F).

5. DEFINITIONS

Term	Definition
Controls	Any device, system, or action that either would likely interrupt the chain of events following an initiating event or would mitigate the undesirable consequences of an Event Scenario. Controls include equipment, design features, specifications, inspection, testing and maintenance routines, operating parameters and other HSE Risk mitigation elements considered in a Risk Analysis.
Critical Mitigation Element (CME)	An independent and reliable device or system that is capable of: 1) preventing initiation of an Event Scenario, 2) preventing loss of Hazardous Material containment, or 3) substantially mitigating one of an Event Scenario's Consequences. Failure of one CME must not prevent another CME from performing its function. CME must be maintained to the reliability level assumed in the risk analysis. A CME reduces an Event Scenario's Probability by one level on the Risk Matrix.
Critical Mitigation System (CMS)	A Risk Mitigation Measure specific to Drilling and Well Servicing which is given credit for reducing the Probability in the Risk Matrix by one level (e.g., Probability Level 5 to Probability Level 4).
EPA Risk Management Plan (RMP)	A U.S. regulatory standard that requires specific types of actions to be taken to prevent accidental releases of certain Hazardous Materials into the atmosphere and to reduce their potential impact.
Event Scenario	A credible and representative potential unplanned event or sequence of events that results in an undesirable consequence. Event Scenarios consist of at least two elements: 1) an initiating event that starts the chain of events, and 2) a consequence that results if the chain of events continues without interruption. Event Scenarios are analyzed for consequence and probability levels to form HSE Risks.
Hazard	A characteristic, activity, arrangement, event, process or substance that has the potential to adversely affect people, the environment, property or revenue.
HSE Risk	The potential of a particular business operation of the Company to give rise to injury or illness to people, environmental impairment, damage to Facilities or other property, or unplanned production interruption. An HSE Risk may relate to operation or maintenance of a Facility, or transportation arranged by the Company of the Company's workers, equipment, products, raw or in-process materials or wastes outside a Facility while in the custody or control of the Company or a Contractor.
Key Mitigation Element (KME)	HSE Risk mitigation elements (equipment, design features, specifications, inspection and maintenance routines, and operating parameters) that are necessary to achieve and maintain the reliability level of Critical Mitigation Elements (CMEs) and Risk Mitigation Measures (RMM), or risk mitigations that are not otherwise claimed as CMEs / RMMs in a Risk Analysis.

Term	Definition
Potential Elevated Consequences (PEC):	Consequences of an Event Scenario that exceed 14 Permanent Incapacitating Injuries or \$1 billion of combined PD/LOR and thereby substantially exceed the threshold for Consequence Level 5.
Property Damage/Loss of Revenue (PD/LOR)	Combined replacement cost for damaged property and the gross loss of revenue to the Company and its joint venture partners and working interest owners in ventures or assets operated by the Company from an Event Scenario
Risk Analysis	A study or review of Event Scenarios that provides qualitative or quantitative consequence and probability estimates that form the Risk Level.
Risk Assessment	The overall process of Hazard identification, Risk Analysis, and Risk Evaluation.
Risk Evaluation	The process of comparing the results of Risk Analysis with risk criteria or standards of acceptability to determine the significance of the risk and to make decisions about future risk mitigation actions.
Risk Level	For an Event Scenario, the Risk Response Category determined by applying the Risk Matrix. When multiple risk response categories are assessed and evaluated, the highest Risk Response Category is reported for the Event Scenario.
Risk Mitigation Measure (RMM)	A group of mitigation measures that are given credit for reducing the Probability in the Risk Matrix by one level (e.g., Probability Level 5 to Probability Level 4) in risk evaluations where the Critical Mitigation Element (CME) methodology cannot be applied.
Risk Register (RR)	The form used by OOG to document C w/PEC, D and E Level Risks. Process Risk Registers are used when CME methodology applies. General Risk Registers are used when CME methodology does not apply. Risk Register types include: Specific: The designation of a Risk Registers when Master, Subordinate or Generic does not apply. Master: When there are multiple Loss of Containment Event Scenarios in the same process block or platform, the Risk Register with the highest Risk Level is designated as the Master. Subordinate: The Risk Registers within the process block or platform whose risks are represented by the Master Risk Register is designated as a Subordinate risk register. Generic: Risk Registers representing substantially similar risks (e.g., third-party excavation of pipelines, risks within the same SFRM Tier Program, etc.) with the same general Mitigation and management plan must be designated as Generic Risk Registers. Generic risk registers may be appropriate at each organizational levels (OPC, OOG and BU).

6. REFERENCES

Document Number	Title
OPC 10:20:80	Confidential Company Information Policy
OOG 10.400.401	Asset Integrity Standard
OOG 10.400.410	Critical Systems and Equipment Assessment Procedure
OOG 60.400.300	HES Risk Management Standard
OOG 60.400.301.03	Developing Event Scenarios Guideline
OOG 60.400.301.06	Controls Systems CME Guideline
OOG 60.400.302	HES Risk Screening Procedure
OOG 60.400.900	HES Audit Procedure

Quantitative Approach (Frequency per year)			OOG HSE RISK MATRIX 60.400.301A Rev 5 Effective 15 March 2020						
↓	Industry History Approach	CMEA Approach							
>10 ⁻²	Unwanted event has occurred in this specific type of operation within this industry using this same technology	No CMEs credited in an Event Scenario	Probability Level	5	B	C	D	E	E
>10 ⁻³ to 10 ⁻²	Unwanted event <u>almost certain to occur</u> in this specific type of operation within this industry using this same technology during the life of the operation	1 CME credited in an Event Scenario		4	B	C	D	D	E
>10 ⁻⁴ to 10 ⁻³	Unwanted event <u>likely to occur</u> in this specific type of operation within this industry using this same technology during the life of the operation	2 CMEs credited in an Event Scenario		3	A	B	C	D	D
>10 ⁻⁶ to 10 ⁻⁴	Unwanted event is <u>unlikely to occur</u> in this specific type of operation within this industry using this same technology during the life of the operation	3 CMEs credited in an Event Scenario		2	A	B	B	C	D
10 ⁻⁶ or less	Unwanted event <u>has not occurred</u> in this specific type of operation within this industry using this same technology	≥ 4 CMEs credited in an Event Scenario		1	A	A	B	B	C
Consequence Categories				1	2	3	4	5	
				Consequences and Consequence Level					
PAF	People At a Facility: Human Injury or Illness			Minor Injuries / illnesses (e.g., First Aid)	Medical Treatment ¹ , Lost work Days	1 Permanent Incapacitating Injury/Illness	2 or 3 Permanent Incapacitating Injuries/Illnesses	>3 Permanent Incapacitating Injuries/Illnesses	
POF	People Outside a Facility: Offsite Human Injury or Illness <u>except</u> Security or Transportation scenarios			Nuisance impacts (e.g., smoke, noise, odor)	Minor Injuries/ Illnesses (e.g., First Aid)	Out-Patient Medical Treatment ¹	In-Patient Medical Treatment	1 Permanent Incapacitating Injury/Illness	
POF-S/T	People Outside a Facility: Offsite Human Injury or Illness related to <u>Security or Transportation</u> scenarios			Minor Injuries/ Illnesses (e.g., First Aid),	Medical Treatment ¹ , Lost Work Days	1 Permanent Incapacitating Injury/Illness	2 or 3 Permanent Incapacitating Injuries/Illnesses	>3 Permanent Incapacitating Injuries/Illnesses	
PD/LOR	Facility Property Damage <u>and</u> Loss of Revenue (<u>excluding</u> Cleanup/Remediation) ²			Up to \$1MM	>\$1MM to \$20MM	> \$20MM to \$50MM	> \$50MM to \$250MM	> \$250MM	

Consequence Categories		1	2	3	4	5
NOP	Non-Owned Property: Damage to Property <u>Not</u> Owned or Operated by OOG or a Contractor (<u>Excluding</u> Cleanup/ Remediation)	Up to \$100M	>\$100M to \$1MM	>\$1MM to \$5MM	>\$5MM to \$20MM	>\$20MM
ECR	Environmental Impairment: Cleanup/Remediation Cost	Up to < \$1MM	> \$1MM to \$5MM	> \$5MM to \$10MM	> \$10MM to \$50MM	> \$50MM or not possible
EHW	Environmental Impairment: Habitat or Wildlife Population	Limited, localized, temporary, and not severe (Up to \$500M)	Localized, short-term with no lasting consequences, slightly severe, and partial restoration /mitigation possible (>\$500M to \$2MM)	Moderate, temporary, moderately severe, and partial restoration/mitigation possible (>\$2MM to \$5MM)	Major, long-term restoration /mitigation possible and severe (>\$5MM)	Extensive, substantial, severe, and restoration not possible, permanent impairment (>\$25MM)
LTO/CR	License To Operate / Community Relations a) Operational impact from external entities; or b) Public reaction; or c) Government or regulatory agency reaction; or d) Emergency actions Note: When applicable to a scenario, utilize the highest of the four sub-categories assessed for the assignment of an LTO/CR consequence level.	a) No operational impacts b) Neighbor inquiries or concern c) No adverse regulatory agency reaction d) Onsite response	a) Potential for suspension of operations at a specific facility lasting several days b) Minor adverse reaction; public inconvenience issues (e.g., wash cars) c) Agency inquiry and reporting d) Incident requires local industries or public responders	a) Potential for suspension of operations at a specific facility lasting a week or more b) Local attention; individual protests or picketing c) Agency investigation and follow-up; community meeting d) Incident could cause shelter-in-place of surrounding community (hours)	a) Operations temporarily suspended in the region pending resolution of issues b) Regional attention; group protests or picketing; blockage to facility access c) Multiple agency inspections or audits d) Incident could cause temporary evacuation (days) of surrounding community	a) Operations revoked in the region or nation permanently or until company corrects structural issues b) National attention; advocacy group lobbying and protests c) Incident directly prompts new government regulation or policy change d) Incident could cause longer-term evacuation of surrounding community

¹ Medical treatment is the treatment beyond first aid of a physical injury or illness at a Medical Center (a public or private hospital, surgery center, or medical clinic).

² For purposes of applying Risk Matrix to operated joint ventures, potential economic effects of an Event Scenario shall be estimated on a gross basis to reflect the venture as a whole, including revenue & costs to the company, joint venture partners or working interest owners, as well as property of Contractors

Appendix B – Risk Response Table

Risk Level	Risk Response	Reviews / Acknowledges	Acknowledges / Signs
A	No further action or risk evaluations required. Business Unit must document identified Hazards, risk evaluations, and Mitigation measures (KME's and CME's)	• BU RMC	• Not required
B	Business Unit must consider feasibility for additional risk evaluations and Mitigation measures (KME's and CME's). Business Unit must document identified Hazards, risk evaluations, and Mitigation measures (KME's and CME's).	• BU RMC	• Not required
C	Business Unit must consider feasibility for additional risk evaluations and Mitigation measures (KME's and CME's). Business Unit must verify Risk Level. Business Units document identified Hazards, risk evaluations and Mitigation measures (KME's and CME's), reasons for actions taken or not taken, and evidence of risk verification.	• BU RMC • BU Operations Manager / VP	• BU General Manager
D & C with PEC	Business Unit must perform appropriate qualitative or, if warranted, quantitative risk analysis to verify the level and identify/evaluate additional or alternative Mitigation measures (KME's and CME's). Business Unit must develop action plans and target dates for implementation of mitigation measures (KME's and CME's) considered to be feasible and determine the target Risk Level. Business Unit must document identified Hazards, risk evaluations and Mitigation measures (KME's and CME's), reasons for actions taken or not taken, and evidence of risk verification. <u>If the HSE Risk cannot be reduced to a lower level within 60 days of identification</u>, Business Unit must report the HSE Risk to OOG HSE Risk Management via the OOG HSE Risk Register, including identified Hazards, risk evaluations, Mitigation measures (KME's and CME's) and implementation status, reasons for actions taken or not taken, target Risk Level, and evidence of verification. Business Units must evaluate the current Risk Level, target Risk Level, previously identified mitigation measures (KME's and CME's) and additional Mitigation opportunities at least annually. OOG HSE Risk Management must validate and report the HSE Risk to OOG Executive Management and OPC HSE <u>within 90 days of identification</u>.	• BU Operations Manager/VP • OOG Dir HSE Risk Mgmt. • OOG VP HSE	• BU General Manager • OOG Presidents
E	In addition to the response elements required for a D Level Risk, concurrence from the OOG President is required continue the affected operation if the HSE Risk cannot be reduced from an E Level to a lower level within 60 days of identification. This concurrence must be renewed at least annually unless the HSE Risk is reduced to a lower level or the affected operation is discontinued. The President's signing of the OOG annual risk report containing the E level risk register provides such annual concurrence.	• BU Operations Manager / VP • OOG Dir HES Risk Mgmt. • OOG VP HSE	• BU General Manager • OOG Presidents
Note: Acknowledgement / signing of risks may be done as a summarized group or report rather than for each individual risk.			

Appendix C – Critical Mitigation Element Analysis (CMEA)

APPLICABILITY

This Appendix is applicable to all Process Hazards Analysis where Critical Mitigation Element Analysis (CMEA) is required - see Section 3.2.4, above.

METHODOLOGY

CMEA is an Oxy-developed approach for assigning a Probability level to Event Scenario Consequences. It utilizes defined, robust risk mitigation measures, termed Critical Mitigation Elements (CMEs) that are credited for reducing the Probability of an Event Scenario Consequence by one level on the OOG Risk Matrix. CMEs are synonymous with strong barriers or “layers of protection”. As such, CMEs are considered Controls that are effective safeguards. Risk assessors identify the number of CMEs that are applicable to an Event Scenario and then, by direct use of the OOG Risk Matrix, determine the Probability level corresponding to the number of credited CMEs.

CMEA was developed to provide more consistent risk analysis results compared with typical qualitative approaches for assigning Probability level to Event Scenario Consequence Scenarios such as the “industry historical approach”. CMEA is considered a simplified and more qualitative form of Layers of Protection Analysis (LOPA) because it only involves identifying the number of applicable layers of protection and does not require any mathematical computation. PHA teams utilizing CMEA need only agree upon the CMEs that are applicable to an Event Scenario and that sufficiently meet the CME characteristics as defined in the following section to assign a Probability level.

CMEA also identifies the elements necessary for a CME to maintain its reliability and effectiveness such as inspections, testing, and maintenance programs, operating and maintenance procedures, and training. These elements are termed Key Mitigation Elements (KMEs). KMEs are also safeguards that are not sufficiently reliable, effective, or independent to be claimed as CMEs and, therefore, are not credited with reduction in Probability level. An example is a process control loop that is not rigorously maintained to achieve the necessary reliability of a CME, is judged ineffective, or is not sufficiently independent of other CMEs credited in the risk analysis.

REQUIREMENTS

1. The following characteristics must be evident to credit a CME in a risk analysis:
 - a) Reliable in providing a low probability of failure such that it achieves at least a ten-fold reduction in frequency of an Event Scenario Consequence, or it functions to lower an Event Scenario's initiating event frequency to 1 in 100 years or less. This typically requires that the component or equipment being credited be registered in the CMMS with an appropriate PM or inspection routine.
 - b) Effective in preventing or significantly mitigating the Event Scenario Consequences when it functions as designed;
 - c) Independent of any other device, system or action that is already being credited as a CME for the same Event Scenario (e.g., over pressure, over fill) and subject to the following conditions:
 - o Only one CME may be credited for a Basic Process Control System (BPCS) loop per control system (e.g., PLC or DCS logic solver) per Event Scenario unless it has been confirmed that the independence criteria is met for crediting two BPCS loops in the same control system as described in OOG 60.400.301.06, Control Systems CME Guideline.
 - o If the initiating event is a failure of a Basic Process Control System (BPCS) loop, then only one CME may be credited for a different BPCS loop in the same control system (i.e., shared PLC or DCS logic solver) provided there is independence between the two loops in terms of the sensing element and final control element.

- o If the initiating event is a human error, then no credit can be taken for any CMEs that would require action by the person who made the error.
 - o If the initiating event is loss of a utility, then CMEs that depend upon that utility cannot be credited unless a reliable utility backup or capacity is available.
 - o Only one CME may be credited for multiple devices or systems in which there is a common cause failure (e.g., independent level sensors that function through independent systems but are subject to the same plugging / failure mechanism could only be credited as one CME).
2. Unless specifically stated otherwise, the Probability of an Event Scenario's Consequences will be reduced by one level on the OOG Risk Matrix for each credited CME.
 3. Initial risk assessments must not credit CMEs for reducing the Consequence Level of Event Scenarios as the criticality ranking of systems and equipment is based upon the highest potential Consequence Level per the requirements of OOG 10.400.411, Critical Systems and Equipment Assessment Procedure and OOG 60.400.303 Facility Technical Information (FTI) Procedure. An exception for when a CME may be credited for reducing the Consequence Level is when it is "passive" and has been verified to provide sufficiently high reliability such that its function to prevent or substantially mitigate Consequences has been judged nearly certain. This exception only applies following rigorous investigation and inspection of the CME's reliability during the risk verification process¹. A passive CME credited for reducing the Consequence Level cannot additionally be credited for reducing the Probability Level of an Event Scenario.
 4. CMEs and their identifier (e.g., PR-6), along with the identified KMEs, must be documented (i.e., listed in PHA or other risk assessment) to allow for verification and validation, as applicable, of their assumed effectiveness in terms of Event Scenario prevention or Mitigation.
 5. Physical CMEs that prevent or mitigate Event Scenarios having Consequences of 3 or higher are designated as "Critical" and those that prevent or mitigate Consequence level 4 or 5 must be managed as HSE Critical Equipment in accordance with OOG 10.400.401 and OOG 10.400.410.

CME TABLES

This Appendix provides a listing of CMEs that are grouped by the following tables:

Table C.1 – Credit for Low Initiating Event Frequency

Table C.2 – Prevent or Substantially Reduce the Probability of a Release

Table C.3 – Prevent Ignition or Explosion - Fire or Explosion Event Scenarios

Table C.4 – Mitigate People Consequences

Table C.5 – Mitigate Environmental Consequences

Table C.6 – Mitigate Financial Consequences

Separate CMEs are provided for SFRM application in Appendix D - Specialized Field Risk Management Program: Wells and Pipeline Event Scenarios and Critical Mitigation Element Descriptions, along with conditions for which Event Scenarios involving SFRM wells and pipelines could be considered non-credible. If the specifications/parameters found in the tables are fully met, the SFRM CMEs may be applied in non-SFRM PHAs.

¹ For cases in which a PHA team believes that a passive CME might be appropriately credited for Consequence reduction, a recommendation should be made for a risk verification to include this evaluation and determination.

Table C.1 – Credit for Low Initiating Event Frequency

This group of CMEs give credit to certain design features, systems, and conditions that would reasonably lower an initiating event frequency to once in 100 years or less.

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
IF-1	Relief Valve High Reliability	This CME applies to Event Scenarios where the initiating event is a relief valve that fails in a full open or substantially open position.	Relief valves in mild service tested every 4 years or less; relief valves in moderate service tested every 3 years or less	1	API 581 Table 7.15 indicates that 10% of all PRD leaks are spurious / large. Per Figure 7.3, leakage failure rates increase with time in service and are less in mild services than more moderate or corrosive services. - Mild services: Crude oil, propane, amine, air, cooling water - Moderate services: Most hydrocarbon liquid and vapor streams, lube and seal oil, process water
IF-2	Innately Safe Design	Removed August 2014. Concept is covered via methodology for developing worst credible Consequences.			
IF-3	Mechanical Integrity – Inherently Safer Materials	This CME credit is applicable where the initiating event is corrosion, erosion, or other material damage mechanism and resistant materials are utilized to prevent or significantly mitigate the damage mechanism.		1	Materials in certain services exhibit a very low degradation rate such that failure of the equipment or piping could be expected to be less than once in 100 years. Use of high alloy piping where normal carbon steel is typically used and the expected corrosion and erosion rate for the service conditions is very low or nil.

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
					Use of A625 CRA or HDPE in a sour crude service in which the corrosion or erosion rate is known to be very low
IF-4	Mechanical Integrity—Increased Design Margin For Detection Before Failure	Removed in Rev. 4 This mitigation is covered by CME PR-7.		1	
IF-5	Mechanical Integrity Design—Self-Announcing	Removed in Rev. 4 Replaced by CME PR-19, Excavation Protection		1	
IF-6	Safety Instrumented System (SIS) / Instrumented Protection System (IPS) High Reliability	This CME applies to Event Scenarios where an Emergency Shutdown (ESD) system, Burner Management System (BMS), or other SIS / IPS failure is the initiating event (e.g., a spurious trip or other failure that activates a shutdown sequence). Note: An IPS is a control or detection system that meets OOG requirements to be credited with Critical Mitigations Elements (CMEs) but does not meet all the technical requirements of a Safety Instrumented System (SIS).	For crediting the low probability of a SIS / IPS failure being an initiating event, the following key requirements of ANSI/ISA 84.00.01 / IEC 61511 must be documented and in place: - SIS / IPS design documentation (Safety Requirements Specification, hardware and application program design, SIL study) - Validation testing of the SIS / IPS to ensure proper operation - Documented and audited program for routine inspection, proof testing,	SIL 1 = 0 SIL 2 = 1 SIL 3 = 2	The failure rate for a SIL 1 Safety Instrumented Function (SIF) could be expected to occur between 1/10 years and 1/100 years (based on a continuous mode operation). This is similar to the initiating event frequencies typically assumed for failure of control loops and other types of failures such that credit is <u>not</u> warranted as an IF CME. More reliable SIL 2 and SIL 3 SIFs would warrant crediting 1 or 2 CMEs, respectively. - Inlet gas high pressure shutdown failure closing the main inlet valve - BMS failure causing continued heating on an HMO heater

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
			calibration and preventive maintenance Operator training on function and operation of SIS / IPS		- Subsea Production Control Systems failure shutting-in production - High Integrity Pressure Protection Systems (HIPPS) spurious trip closing NGL feed to storage
IF-7	Car-Sealed or Locked Valve	This CME applies to Event Scenarios where inadvertent opening / closing of a manual valve due to human error is the initiating event. Inadvertent valve operation at normally unoccupied remote facilities is also considered credible and, therefore, IF-7 could apply to these scenarios as well. The use of CME IF-7 would not be relevant under the following conditions as these scenarios would not be considered credible: - Inadvertent operation of valves that require multiple persons to be involved with its operation (e.g., requires construction of an access platform, uncovering of a buried valve, access to valve located in a pit, or a very large hand-operated valve that would require substantial effort and time to actuate); - Inadvertent operation of a valve for which removal of a slip / spectacle blind / blind flange from the same line would be necessary to cause adverse Consequences as the	A system must be in place (documented list of car seals / locks and periodic checks) to verify that car-seals or locking devices are in place and valves are in the correct position. Car-seals and locking devices must not be easily breakable by hand (i.e. requires a cutting tool to remove), durable so as to remain in place, and secured in a way that prevents a change in valve alignment.	1	The basis for this CME reducing the probability of a valving mistake by an order of magnitude or better is that it: 1) ascribes a higher level of importance to the valve and 2) requires the operator to remove the device and, thereby, be intentional in identifying the correct valve. - The scenario involves opening a gas line valve to #1 unit but an operator might mistakenly try to open the valve to #2 unit located next to it. Since the valves are car-sealed closed and are periodically checked as part of the facility car-seal program, credit can be taken for CME IF-7. - A PHA team identifies a scenario by which a production operator is supposed to open the outlet valve from #1 produced crude tank but inadvertently tries to open the inter-tank connection valve. That valve is shown on the P&ID as a locked-closed valve but there is not a program in place to check the lock periodically. CME IF-7 cannot be claimed to prevent inadvertent

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		needed steps would render this a deliberate, conscious act. This CME cannot be credited if the valve is the same valve for which credit is taken in PR 10.			opening of the inter-tank connection valve.
IF-8	Enabling Condition - Low Time at Risk	This CME applies to scenarios in which conditions exist that could lead to adverse consequences less than 10% of the time. This CME is applied to conditions, whereas CME PC-7 is applied to presence of human receptors less than 10% of the time. Note: An enabling condition is different from double jeopardy events, whereby two or more unrelated failures, each of low probability, must occur simultaneously for the Event Scenario to progress. Such events are generally deemed non-credible due to very low probability of occurrence.	To be credited as a CME, the PHA team must achieve consensus that the enabling condition is present less than 10% of the time and that it is independent of the initiating event. The basis for this consensus must be documented in the PHA worksheet.	1	An enabling condition that exists less than 10% of the time has the effect of reducing the initiating event frequency by an order of magnitude or more (to less than once in 100 years) which is similar to the probability reduction that a physical barrier CME may provide. - A pump is run in a certain configuration about 2 weeks each year to allow for maintenance activities. During these 2 weeks, conditions exist that could lead to a tank overfilling event. This CME could be credited because the time at risk is less than 10% of the time (2 weeks out of 52). - A pressure control failure would normally result in process fluids being safely directed to a surge tank. The surge tank is unavailable for 2 hours each day and during this time, a control failure would result in fluids being directed to an area that could have environmental impacts. Since the time at risk at risk is less than 10% of the time (2 out of 24 hours), a CME credit could be taken for the pressure control failure scenario

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
					having adverse environmental impact.
IF-9	Redundancy	Removed in Rev. 4 Replaced by PR-15, Backup Equipment (Manual or Automatic Activation)			
IF-10	Bull Plug Program	This CME applies to Event Scenarios such as “Vent or Drain Valve Left Open in Error”, or similarly a “Vent or Drain Valve Leaks By” where a bull plug would prevent the scenario from occurring. If the site has a program that assures that bull-plugs are installed and maintained at open-ended lines containing flammable or toxic materials, a CME would be appropriate.	The KMEs for this CME include a documented and audited program for routine monitoring, plug replacement, and LDAR detection (where applicable).	1	The initiating event related to this CME is intended to be a human error leading to an inadvertent or adhoc valve misalignment resulting in LOPC via an open vent or drain.

Table C.2 – Prevent Release or Substantially Reduce the Probability or Consequence of a Release

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
PR-1	Safety Instrumented System (SIS) / Instrumented Protective System (IPS)	This CME applies to SIS and IPS, consisting of sensor(s), logic processor(s) and final element(s) that detect out-of-limit conditions and bring the process to a functionally safe state without human intervention. The Safety Instrumented Functions (SIF) that are implemented within SIS or IPS are separate from Basic Process Control System (BPCS) loops and have an associated Safety Integrity Level (SIL). Note: An IPS is a control or detection system that meets OOG requirements to be credited with CMEs but does not meet all of the technical requirements of a SIS.	The following key requirements of ANSI/ISA 84.00.01 / IEC 61511 must be documented and in place: - SIS / IPS design documentation (Safety Requirements Specification, hardware and application program design, SIL study) - Validation testing of the SIS / IPS to ensure proper operation - Documented and audited program for routine inspection, proof testing, calibration and preventive maintenance - Operator training on function and operation of SIS / IPS	SIL 1 = 1 SIL 2 = 2 SIL 3 = 3	A SIL-1 SIF is synonymous with a CME because of similar expected reliability, requiring a probability of failure on demand (PFD) of $\geq 10^{-2}$ to $< 10^{-1}$ (demand mode of operation). Each additional SIL requires a 10-fold reduction in PFD such that SIL 2 and SIL 3 SIFs are within the range of risk reduction credited to 2 and 3 CMEs, respectively. - High Integrity Pressure Protection Systems (HIPPS) - Loading station ESD on gas detection (executive action, only) - Burner Management Systems (BMS) - Subsea Production Control Systems - Turbine electronic overspeed protection systems
PR-2	Innately Safe Design	Removed in Rev. 4 Concept is covered via methodology for developing worst credible Consequences.			
PR-3	Human Intervention	A CME can be assigned when one or more system(s) or device(s) requires human intervention to take preventative or mitigative action. Such action is typically associated with a BPCS alarm or a system callout prompting operator	Written procedures covering required operator response (may be covered by other means such as a consequences of deviation)	1	A CME credited for human intervention is expected to provide the same level of reliability, effectiveness, and independence as other CMEs. Various factors can influence these CME characteristics such as the amount of time available to detect and

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		intervention. This CME can be credited in addition to an automatic control or ESD system CME if the Human Intervention system is totally independent of these other systems. This CME can be assigned ONLY IF 1) the system or device is monitored sufficiently to detect an upset situation or condition, and 2) personnel have sufficient time to analyze the situation, and take the required action that will prevent or mitigate the Event Scenario. Only 1 human intervention CME credit per Event Scenario is normally allowable. A second CME that involves human action may be credited only when the scenario develops very slowly (i.e. multiple shifts or days) such that different persons would have multiple opportunities to detect the abnormal situation via independent means and have ample time to respond to the situation.	chart / actions to take to correct deviations) • Initial and refresher training (classroom or on-the-job instruction) • Testing via demonstration, simulation, or drills		respond to an abnormal situation, knowing the proper actions to respond, and having the ability to respond. KMEs such as procedures and training help ensure that the level of reliability, effectiveness, and independence are sufficiently strong. • Process vessel low-level alarm that an operator responds to and brings the process back within normal operating limits • During routine operator “rounds” through the plant, an operator observes a level gauge that indicates higher than normal level in a tank and shuts down the inlet flow to the tank to avoid an overfilling situation. • Activation of an emergency shutoff at a truck loading station by the driver attending to the loading operation upon noticing a connection leak.
PR-4	Pressure Relief Devices (PRDs)	This CME applies to potential overpressure scenarios that can be prevented by appropriately sized pressure relief devices. A PRD design basis and relief capacity calculations must be available and verified prior to taking this CME credit.	• PRD design basis and relief capacity calculations • MI program for inspection and testing • Car-sealing or locking device program for designs utilizing block valves on PRD inlets or outlet piping	1	PRDs sized and maintained appropriately to provide an effective means of preventing overpressure events that can lead to equipment rupture or other damage. Typical PFD values for PRDs failing to function are 0.1 to 0.01.

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
					- Pressure Relief Valve (PRV) on compressor discharge - Rupture Disk (RD) on KO drum - Pressure/Vacuum Relief Valve (PVRV) or Pressure/Vacuum Vent (PVV) on crude oil tank
PR-5	Scrubbers and Flares	This CME applies to Event Scenarios involving gases that are reacted, neutralized, or combusted in scrubber systems or flares. It mitigates the Probability that harmful gases will be released directly to atmosphere.	- Flare thermal radiation study and gas dispersion study for potential toxic material releases, as applicable - Documented program for periodic inspection of scrubber or flare systems per applicable MI programs	1	Scrubbers and flares are active mitigation systems that function to prevent the release of gases directly to atmosphere. - Caustic scrubber to neutralize acidic tail gas from a processing system - Flare that combusts gas from episodic pressure relief valve discharges or de-pressuring of a process
PR-6	Basic Process Control System (BPCS) Executive Action	This CME is applicable to BPCS loops (i.e., control functions) within a DCS, PLC, or relay logic hard-wired control system that detect out-of-limit conditions and takes executive (automatic) action to bring the process to a functionally safe state. This CME does not require any human intervention. Note: Some emergency shutdown (ESD) functions and process isolation shutdown valves (SDV) are implemented in a BPCS instead of an independent SIS / IPS (PR-1). As such, they can only be credited one	Documented ITM program for BPCS loops	1	Per ANSI/ISA 84 / IEC 61511, BPCS loops are assigned a risk reduction factor of no more than 10 and in continuous mode of operation, limits the dangerous failure rate to no less than 10^{-5} per hour (1 per 11.4 years), consistent with CME reliability. - Tank high-high level with executive action shutoff function within the DCS control system in response to large in-flow - Temperature switch set to shutdown compressor (PLC) in response to high temperature caused by inlet

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		CME, subject to the independence criteria for control loops. See OOG Control Systems CME Guideline 60.400.301.06 for further guidance on crediting a BPCS loop as a CME.			valve closure that puts compressor into recycle. Pressure control loop within DCS to maintain pressure on column in response to pressure increase from fouling
PR-7	Mechanical Integrity – Increased Design Margin For Detection Before Failure	This CME applies when the initiating event is corrosion, erosion, or other damage mechanisms and there is ample time or opportunity (multiple inspection or test cycles) to detect degradation before an expected failure because the mechanical design is significantly above basic requirements. <u>This CME is not applicable to overpressure Event Scenarios.</u> The actual wall thickness must be at least 50% greater than calculated for mechanical design for the rated operating pressure and temperature or 50% greater than the structural required thickness (whichever is greater). Note: This CME cannot be credited for piping and equipment that is in an MI alarm state (i.e., wall thickness < t_{min} plus 50% of CA); or calculated remaining life less than the established outage frequency or run cycle.	At least one KME must be in place with documented monitoring / testing to ensure that the integrity of the piping or equipment is maintained. Valid KMEs for underground piping are periodic hydrostatic testing and/or periodic smart pigging (In-Line Inspection) Valid KMEs for aboveground piping or equipment that may be subject to general corrosion and erosion damage mechanisms are periodic wall thickness measurements, corrosion probes, and periodic monitoring of corrosion coupons. Valid KMEs for piping or equipment that may be subject to localized damage mechanisms are: Periodic full internal and external visual inspections supplemented by other	1	Increased design margins such as wall thickness significantly greater than required for pressure containment and structural strength provide a greater margin for detection of degradation by the mechanical integrity program prior to LOPC. - Use of schedule 80 pipe where schedule 40 is typically used in a service that has a low expected corrosion / erosion rate and the MI program provides at least two inspections prior to reaching an alarm state. - Use of piping with a Design Factor of 0.5 rather than 0.8 as would normally be acceptable for the piping location.

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
			inspection methods as needed to verify integrity Periodic digital radiography		
PR-8	Secondary Containment Systems (Dikes, Berms)	Removed in Rev. 4 CME is covered by CMEs PC-8, EV-1, and PD-6.		1	
PR-9	Drainage—Directs Hydrocarbons to Safe Location	Removed in Rev. 4 CME is replaced by mitigation CME IE-4			
PR-10	Car-Sealed or Locked Valve	For Event Scenarios where inadvertent opening/closing of a valve is not the initiating event but a car-sealed or locked manual valve would prevent the scenario, this CME would be appropriate. This CME cannot be credited if the valve is the same valve for which credit is taken in IF-7.	A system must be in place (documented list of car seals / locks and periodic checks) to verify that car-seals or locking devices are in place and valves are in the correct position. Car-seals and locking devices must not be easily breakable by hand (i.e. requires a cutting tool to remove), durable so as to remain in place, and secured in a way that prevents a change in valve alignment.	1	The basis for this CME reducing the probability of a valving mistake by an order of magnitude or better is that it: 1) ascribes a higher level of importance to the valve and 2) requires the operator to remove the device and, thereby, be intentional in identifying the correct valve. A high temperature excursion from an upstream process deviation could adversely affect certain downstream equipment but a closed and car-sealed or locked valve on the equipment inlet would prevent potential damage.
PR-11	Check Valve	This CME applies to scenarios for which a substantial reduction in backflow, would prevent or mitigate the event scenario. Check valves may be credited as a CME if slight backflow is tolerable (i.e., tight-	The KMEs for this CME would include a documented and audited program for routine inspection, testing, and preventive maintenance for the check valve.	1	Check valves can substantially prevent backflow of fluids but often have some leak-by when shut, particularly when the service is not clean. Therefore, check valves may be credited when some leak-by is tolerable but not for tight shutoff scenarios. Although two

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		shutoff not required) and if in clean service. Check valves must not be credited as a CME when tight shutoff is required, such as for preventing overpressure due to the potential for leak-by. When two check valves are used in series to increase the reliability, only one CME can be credited.			check valves in series would likely provide better protection against backflow compared to a single check valve, they would not be sufficiently reliable to credit two CMEs due to the potential for common cause failure. - Check valve on discharge of NGL pump to prevent backflow from a parallel pump. - Check valve in a low pressure air system to protect against backflow from a higher pressure system that could potentially overpressure the lower pressure system <u>would not be credited as a CME</u>
PR-12	Concrete Barriers / Bollards and Other Physical Barriers	This CME applies to Event Scenarios involving vehicular loss of control that could impact aboveground equipment or piping, potentially leading to LOPC.	Barriers designed to OOG Engineering Standards, as applicable. Periodic visual inspection	1	- Concrete “jersey barriers” placed around perimeter of a process unit - Steel post barriers / chain installed around a pumping unit - Berm or dike around a storage tank facility that restricts vehicular access
PR-13	Open Vent / Overflow Line	This CME includes passive devices that are designed to relieve material to a safe or contained location. Any intervening block valves must be car-sealed or locked open to ensure functionality. Note: A tank or equipment overpressure / overfilling scenario may be considered non-credible if:	Documentation to support open vent pressure relieving capacity to prevent damage to the tank or equipment if due to blow-by or other potential sources of rapid pressure build-up. Documentation to support overflow line capacity to prevent damage to the tank or equipment due to high	1	Open vents or overflow lines constitute a design feature for preventing high pressure or liquid level from damaging a tank or equipment. - Open vent on a tank to prevent overpressure from a blow-by condition. - Overflow line on a tank to prevent overfilling by discharging the

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		documentation is available to support that: • Documentation exists to support open vent pressure relieving capacity to prevent damage to the tank or equipment if due to blow-by or other potential sources of rapid pressure build-up. • Documentation exists to support overflow line capacity to prevent damage to the tank or equipment due to high inflow or restricted outflow conditions.	inflow or restricted outflow conditions. Documented inspection and maintenance to ensure the vent or line pathway remains unobstructed and meets original design intent. Screens for preventing open vent obstructions Valve car-seal program for any valves in open vents or overflow lines.		material to a safe or contained location.
PR-14	Sampling (Manual or Automatic System)	This CME applies to scenarios involving process or product contamination. It can be credited if sufficient time exists to analyze multiple samples and take corrective actions so as to be considered reliable and effective. This CME cannot be credited in addition to Mechanical Integrity – PR-21, Corrosion Inhibitor System.	QA/QC measures to assure proper sampling techniques and personnel training in place Inspection, Testing, Maintenance program in place for automatic sampling systems	1	Early detection of contamination or improper composition through sampling can effectively protect against damage mechanisms that occur over time if there is sufficient opportunity to verify sampling results and take corrective action. • Daily sampling and testing of process fluid for water content for which high levels could lead to acidic conditions and accelerated corrosion
PR-15	Backup Equipment (Manual or Automatic Activation)	This CME is applicable when backup equipment is available that has sufficient capacity to prevent or mitigate the Consequences of the equipment assumed to have failed and if sufficient time exists to bring the equipment on-stream to prevent or mitigate the Event Scenario.	Each backup equipment (or system) must be inspected, tested, and maintained per applicable asset integrity procedures.	1 or 2	• One CME credit for a full backup instrument air compressor • One CME credit for two similar full backup instrument air compressors (non-diverse) • Two CMEs credited for a full backup instrument air compressor and a full backup nitrogen tank that can be used for instrument air and that has

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		One CME credit can be claimed when there are two or more similar backups. Two CME credits can be claimed if at least one of or more of the backups is diverse in design. CME credit cannot be given when the initiating event involves a common cause failure (e.g., loss of power that would prevent both an operational and backup pump from functioning).			sufficient capacity to allow an orderly shutdown or restoration of instrument air. - Backup electric generator - Uninterruptible Power Supply (UPS)
PR-16	Spare Equipment (Automatic Activation)	Removed in Rev. 4 CME was combined with PR-15.			
PR-17	Double Shaft Seal with Buffer / Barrier Fluid	This CME applies to scenarios in which abnormal operating conditions or rotating equipment operation could lead to seal failure and LOPC. For crediting this CME, the shaft seal must be of double seal design with a buffer or barrier fluid, and be equipped with a pressure or level transmitter, as applicable, with alarm to alert personnel and allow shut down and isolation of the associated equipment before a significant outer seal failure occurs, leading to LOPC. This CME can only be applied when credit for such systems is not taken as an integral element in another CME such as an ESD (CME PR-1) or BPCS (CME PR-6).	The KMEs for this CME include operation of the seal system per applicable API 617 or 682 seal piping plan and periodic ITM of the rotating equipment, including the associated seal system	1	- Double mechanical seal per API 670 Piping Plan 53A on crude oil pump - Tandem dry gas shaft seal per API 617 on centrifugal propane refrigeration compressor
PR-18	Large Capacity Upstream Knock-Out (KO) Vessel	This CME applies to Event Scenarios in which liquids in a gas stream or an abrupt liquid slug could potentially	For slug scenarios, size of the upstream vessel must be supported by calculation to	1	A scrubber vessel that functions to catch large, liquid slugs upstream of a compressor knock-out pot that

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		overwhelm normal equipment knock-out pots (e.g., on a compressor skid) but would likely be captured by a larger capacity upstream K.O. vessel, thereby significantly reducing the potential for damage to downstream equipment and possible LOPC. Note: The CME can be credited if a buffer vessel is provided upstream of equipment that is susceptible to liquid carryover. It does not refer to a conventional FWKO or inlet separator that may generally be in place.	hold the maximum foreseeable liquid slug. A means must be provided to periodically check the liquid level in the upstream KO vessel (e.g., operator rounds or monitoring via level instrumentation, or a high-level alarm) to ensure that the level in the vessel remains low.		functions to prevent carryover of entrained liquids in the gas stream
PR-19	Excavation Protection	This CME applies to prevention of LOPC from an excavation striking an underground pipeline (incidental contact) by use of a physical barrier or reliable warning feature.	QA/QC to assure design and installation will meet the intent of this CME. Aboveground line markers and signage Periodic pipeline surveillance or patrols to ensure line markers and signage have not degraded Excavation permitting process	1	CME Credit is given to buried high-visibility warning tape or mesh as further excavation resulting in an LOPC would require intentional disregard for this common warning practice. - Concrete or steel physical barrier placed above a buried pipeline - Slurry encasement around buried pipeline; red-dyed concrete around buried electrical conduit - Buried high-visibility warning tape or mesh placed a minimum of 12" over the buried pipeline
PR-20	Physical Control or Protection Devices	This CME applies to Event Scenarios for which a physical device can reliably protect against process deviations and mechanical / electrical equipment failure or damage.	Periodic inspection, testing, and maintenance as is applicable to the physical device.	1	- Flow restriction orifice to prevent too high of flow rate - Valve travel stop device to physically prevent a valve from moving beyond a set position

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
					• Pulsation dampener to prevent pressure surge • Mechanical turbine over-speed trip device • Electrical overload protection device to trip a pump motor in event of too high of load and current draw • Slow valve closure time of valves to prevent hydraulic hammer in liquid services • Vibration switch on cooling tower fan to detect high vibration and shutdown fan motor • Centrifugal compressor vibration / axial displacement monitoring system
PR-21	Mechanical Integrity – Corrosion Inhibitor System	This CME is applicable to Event Scenarios where the initiating event is internal corrosion and a corrosion inhibitor system is in place to prevent or significantly mitigate the potential for internal corrosion.	• Monitoring of corrosion inhibitor effectiveness (residual concentration, corrosion coupons, etc.) • Chemical provider usage monitoring program • ITM program in place per applicable OOG AI/MI procedure.	1	A corrosion inhibitor system, when effectively applied and monitored, provides a robust means of reducing the probability of internal corrosion that could lead to a leak. • Pipeline corrosion inhibitor addition system
PR-22	Mechanical Integrity – Cathodic Protection (CP) System	This CME applies to Event Scenarios involving potential external corrosion of underground piping and pipelines.	The required KME for CP systems include inspection, testing, and maintenance per NACE SP 0169.	1	A CP system can provide an effective means of protection against leaks in buried piping and pipelines when properly maintained and inspected. Therefore, this CME can be credited to reduce the Probability of a leak.

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
					Example: CP system applied to an underground pipeline that is coated with fusion-bonded epoxy (FBE) for external protection
PR-23	Secondary Verification of Process Isolation Barriers	This CME applies to Event Scenarios involving a potential inadvertent valve opening or machine operation that could result in personnel exposure to high pressure or corrosive / toxic / flammable material, or equipment damage due to high pressure or other undesired factors. This CME cannot be credited with other CMEs involving Human Intervention such as IF-7, IF-10, or PR-3. In addition, the CME cannot be credited if an inadvertent valve operation is postulated to take place after the system has been verified by the secondary personnel.	The KME for this CME includes a documented and audited isolation / permitting control of work program, incorporating clear instructions for secondary personnel checks (i.e. a documented valve or other isolation barrier status verification by another person).	1	Certain activities in which an error could be made that leads to an LOPC can be substantially mitigated by a structured system in which multiple persons serve as checks for the activity's tasks and the tasks are clearly defined, documented, and can be audited. Scraper (pig) launching / receiving procedure with specific task steps and documented verification by a qualified second operator apart from the primary operator who is performing the tasks

Table C.3 – Fire or Explosion Event Scenarios - Prevention of Ignition and Fire or Explosion

These CMEs reduce the Probability of ignition that can lead to a fire or explosion.

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
IE-1	Low Probability of Ignition	This CME is applicable to flammable material releases when there is a low probability of ignition due to location of ignition sources or prevailing wind direction in relation to a flammable material release location.	- Separation distances between potential release locations and ignition sources meet facility layout and spacing standards (OOG EG-10-400-501 and 502) - When low probability of ignition is ascribed to prevailing wind direction, a wind rose is available to substantiate prevention of ignition 90% of the time during a year - Work Control systems (permit to work; hot work permit) are enforced for transient ignition source activities	1	Within process plants, Quantitative Risk Analysis (QRA) data sets typically assign an ignition probability of less than 10% for releases of 2-inch / 50 mm hole size and 1% for 0.5-inch / 12.5 mm hole size. These probability values for release sizes typically assumed within facilities are comparable to the Probability reduction ascribed to CMEs. - Flammable material release in a remote part of a plant, tank farm, or tank battery where sources of ignition such as heaters, engines, and electrical switchgear are not likely to be present - Flammable material release where the prevailing wind direction is towards potential ignition sources no more than 10% of the time during a year - Flammable gas detector for building air intake to prevent ingress of flammable materials into the building and possible ignition - Pipeline or piping system designed such that potential failure from over-pressure is likely to occur at a weaker section located away from people and process equipment
IE-2	Explosion Protective Spacing/Layout	Removed in Rev. 4 CME concept is covered within facility siting studies.		1	

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
IE-3	Tank Gas Blanketing	This CME is a design measure to prevent an explosion by eliminating oxygen within the vapor space of tanks containing hydrocarbons. The blanket must be an engineered system using inert or other gases (typically methane) that will keep the vapor space outside the flammable range under all tank operating conditions.	Appropriate KMEs must be applied to ensure that the blanketing system is properly designed, operated, and maintained on a documented basis .	1	The Event Scenario that relates to this CME is intended to be air (oxygen) ingress into a tank due to vacuum condition or a tank breach condition.
IE-4	Drainage – Directs Flammable Materials to Safe Location	This CME applies to LOPC events involving a flammable material that could pool around equipment and, if ignited, could directly affect the equipment or escalate to other equipment.	The KME for this CME includes a documented program for periodic inspection of the drainage system to ensure that there is sufficient ground slope for area drainage away from equipment and that impoundment basins, if provided, are not degraded.	1	LOPA typically assigns drainage systems a PFD of 10^{-2} or less which is on the conservative end of the range for CME reliability: - Sloped concrete grading and drainage channel to direct spill from butane storage tank to an impounding basin - Ground grading and curbs that would direct a hydrocarbon spill from a process or storage vessel away from equipment and into an open area.

Table C.4 –Mitigate People Consequences

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
PC-1	Safe Shelter-in-Place or Toxic Gas Refuge	This CME applies when building occupants can be protected against a toxic gas release by measures that limit exposure from infiltration or by air pressurization systems that protect occupants until a toxic cloud has passed, or until occupants can escape via emergency escape respirators or breathing air. If escape is necessary, then reliable emergency routes must also be available.	Air infiltration / exposure time estimation to support adequacy of protection for personnel against credible toxic release scenarios Training of personnel on Sheltering-In-Place and escape / evacuation procedures. Inspection, testing, and maintenance program for positive pressurization system	1	Appropriately designed facilities for preventing ingress of toxic gases into a sheltering or work location, or supplied breathing air, is an effective means of protection. Due to potential time constraints associated with these systems, provision for emergency escape or evacuation may be necessary. • Designated Shelter-In-Place room within a building that is sufficiently tight against toxic gas infiltration • Control room positive pressurization system • Fixed breathing air system / Self-Contained Breathing Apparatus
PC-2	Fire and Blast Walls or Hardened Buildings	This passive mitigation CME group applies to devices that are designed and maintained to provide a level of protection for people from the fire or blast effects of an ignited release, or if the system provides time for people to evacuate to a location that protects them from significant injuries. This CME cannot be credited with PC-5, Immediate Alert and Escape Route due to potential for rapidly developing scenarios.	Design documentation for fire wall, blast wall, or hardened building features that support adequacy of protection based on postulated fire and explosion scenarios from facility siting or similar studies. Documented inspection / maintenance program for measures credited must be in place.	1	For protection of people, these passive controls primarily serve to reduce the vulnerability of building occupants. Where facility siting studies are available that include predicted building damage levels, this CME could be applied to relevant explosion hazard scenarios. Occupied buildings that have a building damage level of 2.5 or less could be credited with the CME based on an occupant vulnerability of less than 0.1. Note: More rigorous risk analysis means would typically be employed for these types of scenarios. • Fire wall that would provide protection against a postulated fire scenario for

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
					occupants, and the escape route is sufficiently shielded for evacuation to a safe location. • Blast-resistant building for protecting remote I/O control system equipment from possible fire and explosion effects
PC-3	Fireproofing (Passive Fire Protection)	This passive mitigation CME applies to materials that are designed and maintained to reduce the rate of heat input to vessels, piping and structures in order to delay or prevent Consequences (e.g., to provide additional time for depressurizing a vessel or firefighting). For potential fire Consequences to people, this CME can be credited for reducing the probability of fire exposure if enough delay is provided to allow people to evacuate to a location that protects them from significant injuries.	Documented inspection / maintenance program for PFP must be in place. In addition, documentation should be available to support appropriate design for fire proofing based on postulated LOPC scenarios (e.g. fire exposure envelopes conforming with API 2218 Fireproofing Practices, and EG-308 Fireproofing).	1	With routine inspection / maintenance, passive fire protection (PFP) provides sufficient reliability to be a CME. • Dense concrete fire-protection on gas processing unit structural steel with a 2-hour UL 1709 fire resistance rating • Intumescent mastic applied to a storage vessel to prevent escalation of a fire and that provides sufficient time for people to evacuate the area.
PC-4	Active Fire Protection and Fire Suppression Systems	An active fire protection system can reduce the rate of heat input to vessels, piping and structures and potentially delay or prevent Consequences (e.g., to provide additional time for depressurizing a vessel or firefighting) or can protect people from significant injuries during their escape. This CME can be credited if enough delay or exit protection is provided to allow people to evacuate to a location that protects them from significant injuries.	Appropriate documented, design, maintenance, testing, and inspection program for the active fire protection system in accordance with consensus NFPA standards must be in place.	1	The Event Scenario that relates to this CME is intended to be a flammable material release that could affect a fixed receptor such as an occupied building. • Emergency Response Team action to extinguish an occupied building fire • Water spray system covering a propane storage tank to reduce the probability of an area fire escalating to a BLEVE event that could adversely affect control room operators

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		This CME applies to fire hazard Event Scenarios for which the system suppresses a fire to protect building occupants or that prevents escalation of a fire that could put people in harm's way.			Automatic fire sprinkler system to control or extinguish an interior occupied building fire.
PC-5	Immediate Alert and Escape Route	This CME applies to fire and toxic material release Event Scenarios in which people would be immediately alerted of the fire or release and are located in a building or otherwise protected from immediate harm, and they have a protected escape route (e.g., building or structure to mitigate thermal radiation exposure). Toxic and flammable scenarios must be evaluated separately. This CME does not apply to explosion / flash fire events and it cannot be credited in addition to PC-6, Supplied Air Systems.	Typical documentation to support the CME includes the related dispersion modeling report, or a facility siting study. ITM program for the system used to provide detection and alerting of people	1	As the CME involves immediate notification and escape, the circumstances allowing the CME credit should be clearly supported in the PHA worksheet. Such support should include clear consensus among the PHA team that the personnel are located sufficiently away from the LOPC source and will have ample time to escape through a protected route. - Plant fire alarm system that alerts occupants of a maintenance shop located near a process area of a fire or potential fire in the process units, and the building structure is sufficiently large to shield the occupants from hazardous thermal radiation levels that they would otherwise be exposed while evacuating the building. - A gas detector at a tank battery that provides warning to an operator of potentially high H₂S gas concentration in the area and the operator can safely move cross-wind or upwind of the release source.
PC-6	Supplied Air Systems	This CME applies to an Event Scenario where an operator or maintenance person must operate or maintain equipment such as a	Documented and audited program that ensures proper respiratory protection equipment and		A CME can be taken for positive pressure SCBA if the facility has a documented, audited program that assures that SCBA

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		valve where their actions could potentially release toxic gases but robust protection is provided by a supplied breathing air system or SCBA that is required for the activity. This CME also includes escape SCBA for which sufficient time exists for donning the apparatus and for egress / escape from the This CME cannot be credited in addition to PC-5 Alert and Escape Route.	usage each time that work is performed involving these types of activities. Respiratory protection program that complies with OOG requirements.		is used each time work is performed that could lead to this scenario.
PC-7	Personnel Receptor – Low Time at Risk	This CME can be credited for a lower Probability of potential injury if personnel rarely (< 10% of the time) would be in the area affected by the Event Scenario and if persons who might be present are not associated with the cause and mitigation of the Event Scenario.	To be credited as a CME, the PHA team must achieve consensus that the time presence of potentially affected personnel is <10% of time. The basis for this consensus must be documented in the PHA worksheet.	1	For Event Scenarios in which the cause and mitigation of the scenario is unrelated to persons who might be present (i.e., random event), a low presence factor of 10% or less sufficiently reduces the likelihood of an injury consistent with that of a CME. • Process controls initiated excursion at a flare or a tank farm. If personnel are in these areas of a facility on a limited basis, then a CME PC-7 credit can be applied to account for lower probability of personnel impact. • Potential for injury from VRU failure while operator is performing daily surveillance checks at a tank battery. Operator is at the location about 30 minutes per day on average (2% of time) such that CME PC-7 would be credited for reducing the probability of adverse impacts. • Harmful impact scenario to personnel performing a routine operational or

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
					maintenance surveillance round under a buddy system (at least 2 operators in attendance) in a remote area of a manned facility, with a time presence of < 2.4 hours per day. Although the Consequence Level may be assigned as at least 4, the PC-7 CME would still be applicable to reduce the Probability level.
PC-8	Secondary Containment Systems (Dikes, Berms)	This CME applies to LOPC events and mitigates spills and releases of material from spreading beyond the primary containment that is assumed to fail in the risk analysis.	The KMEs for this CME include – • A documented program for periodic inspection of the secondary containment system that will detect potential degradation that could lead to greater Consequences than assumed in the risk analysis. • Calculations to support that the containment has sufficient capacity to handle the anticipated release volume, including rainfall and firewater usage, as applicable.	1	LOPA typically assigns a PFD for secondary containment of 10^{-2} or less which is on the conservative end of the range for CME reliability. • Coated-concrete containment dike around an acid storage tank • Earthen berm around a production tank battery • Concrete curbing around an electrical transformer • Double-walled tank for storing fuel • Berms between oil tanks in a tank farm area to minimize potential for a spill spreading to other tanks

Table C.5 –Mitigate Environmental Consequences

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
EV-1	Secondary Containment Systems (Dikes, Berms)	This CME applies to LOPC events and mitigates spills and releases of material from spreading beyond the primary containment that is assumed to fail in the risk analysis.	The KMEs for this CME include – • A documented program for periodic inspection of the secondary containment system that will detect potential degradation that could lead to greater Consequences than assumed in the risk analysis. • Calculations to support that the containment has sufficient capacity to handle the anticipated release volume, including rainfall and firewater usage, as applicable.	1	LOPA typically assigns secondary containment a PFD of 10^{-2} or less which is on the conservative end of the range for CME reliability. • Coated-concrete containment dike around an acid storage tank • Earthen berm around a production tank battery • Concrete curbing around an electrical transformer • Double-walled tank for storing fuel • Berms between oil tanks in a tank farm area to minimize potential for a spill spreading to other tanks
EV-2	Environmental Receptor – Low Time at Risk	This CME will be credited if data can demonstrate the condition that could result in the Event Scenario's Consequences exists less than 10% of the time (i.e., the event would have to occur at the same time that another condition was in effect but that condition is only applicable less than 10% of the time).	To be credited as a CME, the PHA team must achieve consensus that the potentially affected receptor is vulnerable <10% of time. The basis for this consensus must be documented in the PHA worksheet.	1	Example: A tank overflows during the <u>limited window of time the tank dike's storm drain is open (which is open less than 10% of the time)</u> allowing a spill to reach a water receptor.
EV-3	Human Intervention – Environmental	This CME is applicable for ERP / SPCC Plan with human intervention to prevent or mitigate an Event Scenario. A CME can also be credited for emergency response / spill plans if the spill will be detected, equipment and	To be credited as a CME, the PHA team must achieve consensus on the postulated personnel reaction time allowing the release to be contained in respect to the postulated release size and	1	

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		manpower are always present ($\geq 90\%$ of the time or there is sufficient time for personnel to respond to a spill if notified) and the scenario will develop slowly enough for the spill/release to be intercepted before it reaches the receptor. An example is an oil spill released that will take an hour to reach a stream but there is a very high confidence that the event would be detected and booms would be laid in less than an hour to prevent any significant amount of oil from reaching the stream). This CME cannot be used in conjunction with any other CME that requires human intervention.	the time required to reach the environmental receptor. The basis for this consensus must be documented in the PHA worksheet.		

Table C.6 – Mitigate Financial Consequences

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
PD-1	Layout/Spacing/Separation Distance	Removed in Rev. 4 Concept is covered via engineering standards and facility siting studies			
PD-2	Fire and Blast Walls	This passive mitigation CME group applies to devices that are designed and maintained to provide a level of protection for equipment from fire or blast effects of an ignited release. CME credit is applicable if it can be demonstrated and documented (typically a consequence analysis) that PD/LOR will be substantially reduced because of the devices.	Fire wall design rating (hours) supported by consequence analysis of postulated fire scenarios Blast wall design rating (overpressure) supported by blast modeling of postulated explosion scenarios.	1	LOPA typically assigns fire and blast walls a PFD of 10^{-2} or less which is on the conservative end of the range for CME reliability. - Blast wall constructed to reduce the potential for a postulated blast from damaging critical equipment - Fire wall on offshore platform to shield control equipment from high thermal radiation level
PD-3	Fireproofing (Passive Fire Protection)	This passive mitigation CME applies to Event Scenarios involving potential pool fires and jet fires that could affect assets in terms of property damage or associated loss of revenue due to associated downtime. A CME credit can be taken if it can be demonstrated and documented that PD/LOR will be substantially reduced because of these materials.	Fire-resistance rating (hours) supported by fire hazards analysis per API 2218, giving consideration to fire duration / intensity, escalation (potential collapse) and recovery / replacement time of plant / equipment.	1	Fireproofing functions to reduce the rate of heat input to vessels, piping and structures in order to delay or prevent Consequences (e.g., to provide additional time for depressurizing a vessel or firefighting). - Critical vessel supporting structures protected with concrete fireproofing - Liquid hydrocarbon vessels protected with an epoxy intumescent coating - Piping protected with fire-resistant flexible endothermic matting
PD-4	Active Fire Protection and Fire Suppression Systems	This CME applies to Event Scenarios involving potential fires that could	Documented design basis and design of system	1	Active fire protection and fire suppression systems function to

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
		affect assets in terms of property damage or associated loss of revenue due to associated downtime.	- ITM program for the fire protection or suppression system - Trained and capable Emergency Response Team with periodic drills, as applicable - Sufficient time for fire detection, response, and protection / suppression actions to be effective		extinguish fires, prevent escalation by reducing the rate of heat input to equipment and structures, or provide additional time for system depressurization and firefighting response. - Fire or flammable gas detection system with executive action or Emergency Response Team actions - Water spray system covering a propane storage tank to reduce the probability of an area fire escalating to a BLEVE event that could result in extensive PD/LOR - Automatic fire sprinkler system to extinguish a warehouse fire - CO₂ or clean agent fire suppression system to mitigate an I/O room fire
PD-5	Process Isolation Valves	Removed in Rev. 4 Mitigation is covered by PR-1, PR-3, and PR-6			
PD-6	Secondary Containment Systems (Dikes, Berms)	This CME applies to LOPC events and mitigates spills and releases of material from spreading beyond the primary containment that is assumed to fail in the risk analysis.	The KMEs for this CME include – A documented program for periodic inspection of the secondary containment system that will detect potential degradation that could lead to greater Consequences than assumed in the risk analysis.	1	LOPA typically assigns secondary containment a PFD of 10⁻² or less which is on the conservative end of the range for CME reliability. - Coated-concrete containment dike around an acid storage tank - Earthen berm around a production tank battery - Concrete curbing around an electrical transformer - Double-walled tank for storing fuel

CME ID	CME Description	Applicability	KME Requirements	CME Credits	Application Basis / Examples
			• Calculations to support that the containment has sufficient capacity to handle the anticipated release volume, including rainfall and firewater usage, as applicable.		• Berms between oil tanks in a tank farm area to minimize potential for a spill spreading to other tanks

Appendix D – Specialized Field Risk Management Program: Wells and Pipeline Event Scenario and Critical Mitigation Element Descriptions

Table D.1 – Non-Credible SFRM Well Scenarios

SFRM WELLS	
LOPC from Production Casing to Surface Casing	Overpressure and corrosion Event Scenarios for Production Casing LOPC to the Surface Casing may be considered non-credible if the following criteria are met: • Cement is run the entire length of the annulus between Production and Surface Casings • Bond log is run to assure successful cement job to surface on Production Casing String • Bradenhead monitors pressure (designed per Control System CME Guideline) in annulus between Production Casing and Surface Casing, alarms and requires operator response • External casing packer in the annulus between the Production and Surface Casing is provided to act as an additional barrier for gas flow (this specific requirement only applicable to new drills after 01/01/2014)
SFRM Size LOPC Due to Tubing Failure From Rod based Wear	An SFRM type LOPC Event Scenario due to wear caused by a Beam pump's rod may be considered non-credible if the following criteria are met: • New top tubing joint is installed each time the tubing is pulled • No unusual situation is present that would suggest that a leak from this Event Scenario would be more than a small leak Note: Rod wear typically occurs within the well bore structure, which should minimize potential for a surface release.
Wellhead Corrosion	An SFRM type LOPC Event Scenario due to corrosion of the wellhead itself may be considered non-credible if: • The wellhead meets applicable SFRM design requirements • All wellhead fittings are of flanged construction • No unusual situation is present that would suggest that a leak from this Event Scenario would be more than a small leak Note: Due to robustness of this equipment and design integrity, low probability of occurrence deemed for a large bore SFRM type release resulting from equipment corrosion.

Table D.2 – SFRM Well Critical Mitigation Elements (CMEs)

SFRM WELLS CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
Three String Casing Design	SFRM -1	1	Overpressure or Corrosion LOPC of Production Casing	-Intermediate string has to be rated for same design as Production Casing - QA/QC measures to assure proper installation and ongoing reliability -Casing Integrity Log run initially and every 10 years or sooner, depending on the condition of the casing	Ongoing reliability to be addressed by AI/MI program
Production Casing Full Liner Installation on Non-Failed Well (run to the Surface)	SFRM -2	1	Overpressure or Corrosion LOPC of Production Casing	-Requires liner installed to depth that would prevent a surface blowout -QA/QC measures to assure proper installation and ongoing reliability - Casing Integrity Log run initially and every 10 years or sooner, depending on the condition of the casing - Validate the integrity of the existing production casing prior to installation of the liner	Depth below where surface blowout would occur must be based on a documented geological review Ongoing reliability to be addressed by AI/MI program
Production Casing External Coating Applied Through Zones Where Corrosion Is Expected To Be An Issue	SFRM -3	1	Applies to external corrosion LOPC Event Scenarios when the production casing passes through zones where external corrosion is expected to be more aggressive than normal	- Coating system must be verified to be appropriate for expected corrosion mechanisms - QA/QC program for coating application - QA/QC program for inspection before commissioning well - Appropriate procedures for handling and placing in the well	This CME, if applicable, may be credited in addition to Cathodic Protection for SFRM Wells
Bradenhead Pressure Alarm and Operator Response	SFRM -4	1 (PR-3)	Corrosion LOPC between Production Casing and Surface Casing TA well LOPC - fluid in annular space	-Requires alarm with operator response -QA/QC measures to assure proper installation and ongoing reliability of alarm system	-Operator response can only be used one time for any Event Scenario

SFRM WELLS CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
H2S Detection Around Well Pad Interlocked to ESD Valve on Flowline coupled with Operator Alarm	SFRM -5	1 (PR-6)	Overpressure LOPC from tubing or well hookup piping OR Corrosion LOPC for injection wells only	- Requires automated valve closure coupled with alarm which is either on separate shutdown/control system or hardwired - QA/QC measures to assure proper installation and ongoing reliability of alarm/control systems	- Alarm/shutdown must be designed per Control System CME Guideline - Cannot be credited in addition to other shutdowns unless independent.
SFRM Vehicle Impact Barriers	SFRM -6	2 (PR-12)	Vehicle Impacts to above ground piping having elevated risk: Barriers engineered for this purpose would have a very high probability of preventing damage that would cause LOPC	Barriers designed to OOG Engineering Standard. Barriers are required for instances where there is an identified elevated probability that the wellhead, tree assembly and hook-up piping/equipment could be damaged by an impact resulting in an SFRM size LOPC.	Reference 2013 SFRM Well Vehicle Impact Protection Study by Oxy Permian Risk
Chemical Treatment System	SFRM -7	1 (PR-21)	Internal Corrosion LOPC from Production Casing Leak or Internal Corrosion on Well hookup Piping	-Chemical treatment system for well bore tubulars which also benefits downstream equipment. This is operating practice for well bore protection and serves as a prevention of internal corrosion -Chemical treatment application are different for different type wells, ensure corrosion coupon monitoring and other adequate KMEs are followed for proper monitoring of chemical treatment -QA/QC measure to ensure wellhead, tree, tubulars and associated instrumentation are properly maintained in MI program	
Stainless Steel on Well Hookup Piping and Components	SFRM -8	1 (IF-3)	Well hookup failure due to internal corrosion	- Metallurgy must be verified to be appropriate for expected corrosion mechanisms	Part of generic well design standard

SFRM WELLS CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
Cathodic Protection for SFRM Wells	SFRM -9	1 (PR-22)	Protects Surface and Production Casings against external corrosion	-QA/QC measures to assure proper installation and ongoing reliability of cathodic protection system	This CME, if applicable, may be credited in addition to Production Casing External Coating Applied Through Zones Where Corrosion Is Expected To Be An Issue
ESP Low Amperage Alarm/ Shutdown	SFRM -10	1 (PR-6)	Overpressure of the Well Hook-up above surface based on ESP blocked outlet.	-Requires automated ESP shutdown -QA/QC measures to assure proper installation and ongoing reliability of alarm/control systems	-ESP low amperage alarm/shutdown must be designed per Control System CME Guideline
High Flowline Pressure Device Interlocked to Shut Down ESP	SFRM -11	1 (PR-6)	Overpressure LOPC from tubing or well hookup piping based on ESP blocked outlet	-Requires automatic shutdown which is either on separate shutdown/control system or hardwired -QA/QC measures to assure proper installation and ongoing reliability of alarm/control systems	- ESP Alarm/shutdown must be designed per Control System CME Guideline - Cannot be credited in addition to ESP low amp shutdown unless independent
Position Switch on Block Valve(s) for ESP Shutdown & Startup Permissive	SFRM -12	1 (PR-6)	Overpressure of the Well Hook-up above surface based on ESP blocked outlet	-Requires automated ESP shutdown and open valve startup permissive -QA/QC measures to assure proper installation and ongoing reliability of alarm/control systems	- Limit Switch on block valve must be designed per Control System CME Guideline
Beam Unit High Pressure Stuffing Box with High Pressure Shutdown (e.g. set at 200 psig)	SFRM -13	1 (PR-6)	Overpressure LOPC of Beam Unit Well Hook Up	-Requires reliable leak/stuffing failure alarm, motor shutdown and operator response -QA/QC measures to assure proper installation and ongoing reliability	-High pressure alarm/shutdown must be designed per Control System CME Guideline -With automated motor shutdown, operator response will not count against the one human intervention limit

SFRM WELLS CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
Beam Unit Stuffing Box Failure Applied As a Pressure Relief Device (note this protection is applicable for the Beam Unit Well Hook Up, not the flow line)	SFRM -14	1 (PR-4)	Overpressure caused by pressure from rod pump	-Requires: a) documentation that stuffing box will fail below the pressure components' rating by a minimum of 500psig b) 0.5 inch ROE from stuffing box failure will not impact public receptors at the SFRM thresholds (0.5-inch assumed to be equivalent hole size for stuffing box packing failure) -QA/QC measures to assure proper installation and ongoing reliability of stuffing box's failure pressure (e.g., MOC of design/equipment changes)	ROE is Radius Of Exposure corresponding to the SFRM screening thresholds of OOG 60.400.302 HES Risk Screening.
Beam Unit Well Hookup High Pressure Device to Shutdown Unit	SFRM -15	1 (PR-6)	Overpressure LOPC of Beam Unit Well Hook Up	-Automatic shutdown which is hardwired (or on other independent system) and independent of the main Beam Unit PLC, as stated in design and operating requirements - QA/QC measures to assure proper installation and ongoing reliability of alarm/control systems	Beam Unit Motor Alarm/shutdown must be designed per Control System CME Guideline
Orientation of Beam Pump To Protect Wellhead Equipment	SFRM -16	2 (PR-12)	LOPC from vehicle impacts Requires that beam pump be oriented so that it would protect the wellhead from the direction of potential vehicle traffic	-MOC if orientation is changed or road(s) are modified	Risk engineering review of potential vehicle impacts required. This would be the same CME credit allowed for SFRM Vehicle Impact Barriers

SFRM WELLS CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
Operator Intervention For Slowly Developing Situations With Distinctive Alarm(s)	SFRM -17	1 (PR-3)	SFRM LOPC caused by various initiating events when: • event scenario slowly develops • condition is a distinctive process deviation (e.g., excess CO2 in gas) or high priority alarm that doesn't require analysis to determine condition • ample intervention time is available between deviation/alarm and SFRM release	-QA/QC measures to assure proper installation and ongoing reliability of deviation/alarms that allow ample time to react	A distinctive deviation/alarm is one that will immediately come to the attention of the operator without requiring data analysis. This CME may be taken in addition to a dedicated alarm for acute situations requiring human intervention Applicable D&O requirements include those related to weekly surveillance by Production Tech, and Well Analyst.
Production Casing Pressure Monitoring Device with Alarm and Shutdown of Choke Valve / ESD (Applies to Injectors and Flowing Wells with packers, only)	SFRM -18	1 (PR-6) Exec. Action	Overpressure of Production Tubing resulting in Tubing Failure	-Independent shutdown (hard wired or separate control / shutdown system) of injector / flowing well choke / ESD valve under high pressure condition -QA/QC measures to assure proper installation and ongoing reliability of alarm/control system(s)	-Alarm and control systems must be designed per Control System CME Guideline
Production Casing Pressure Alarm and Operator Response (Applies to Injectors, Flowing Wells only – both with packer fluid, and Temporarily Abandoned wells)	SFRM -19	1 (PR-3)	Corrosion LOPC between Production Tubing and Production Casing (i.e. a tubing leak/failure from TA well mechanism)	-Requires alarm with operator response -QA/QC measures to assure proper installation and ongoing reliability of alarm system	-Operator response can only be used one time for any Event Scenario

SFRM WELLS CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
Inhibited Water Solution (Packer Fluid) in Production Casing (Annulus between Production Casing and Production Tubing) – (Applies to Injectors and Flowing Wells only)	SFRM -20	1 PR-21	Corrosion LOPC caused by internal corrosion of Production Casing	-QA/QC measures to assure proper selection of inhibitor -Credit annual casing integrity test required by TRRC and NMOCD -Inhibitor water is checked and topped-off annually	
Lined/Coated Tubing Used (fiberglass or plastic coated) and Inhibited Water Solution (Packer fluid) on outside of the Tubing (Applies to Injectors only and Flowing Wells – both with Packer Fluid)	SFRM -21	1 (IF-3)	Corrosion LOPC between Production Tubing and Production Casing (i.e. a tubing leak/failure)	- Lining/coating system must be verified to be appropriate for expected corrosion mechanisms - QA/QC measures to assure proper installation and ongoing reliability - Packer fluid on outside of tubing part of standard operating practice	
ESP Well Hookup Piping Inherently Safe Design	SFRM -22	1	Overpressure of the Well Hook-up above surface based on ESP blocked outlet	- All wellhead fitting are of flanged construction - The piping shall be constructed with a user defined construction type design factor of 0.5 using the maximum anticipated ESP dead head pressure as the design pressure. - Flanges and valves shall be designed to meet or exceed the design pressure.	With adequately rated pipe, overpressure of the pipe due to pump deadhead may not be feasible.
Production Casing Robust Design (applies to Injection Wells only)	SFRM -23	1	Overpressure of the Production Casing from Injection Pressure (e.g. due to tubing failure)	-The current production casing burst pressure (as determined by USIT measurements and Barlow equation) should exceed the maximum credible injection pressure from supplied equipment by a minimum of 500 psig. -QA/QC measures to assure proper installation and ongoing reliability -Casing Integrity Log run initially and every 10 years or sooner, depending on the condition of the casing	Ongoing reliability to be addressed by AI/MI program

SFRM WELLS CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
Production Casing Robust Design (applies to Production Wells only)	SFRM -24	1	Overpressure of the Production Casing from Well Shut-in Pressure (e.g. due to well shut-in condition)	-The current production casing burst pressure (as determined by USIT measurements and Barlow equation) should exceed the maximum credible surface well shut-in pressure by a minimum of 500 psig -QA/QC measures to assure proper installation and ongoing reliability -Casing Integrity Log run initially and every 10 years or sooner, depending on the condition of the casing	Ongoing reliability to be addressed by AI/MI program

Table D.3 – Non-Credible SFRM Pipeline Scenarios

SFRM PIPELINES	
Non-Credible Event Scenario	For purposes of SFRM risk evaluation, Event Scenarios for overpressure / internal & external corrosion may be considered non-credible if the following criteria are met: • Materials of construction meet SFRM Design Standards for its service and verified by QA/QC program • Pipe is seamless with 100% RT or UT of shop and field welded joints • Pipeline system is rated for 1.5 times the maximum source pressure (e.g., well formation, pump, or compressor) • Hydrostatic test conducted per applicable ASME code for the pipeline system Mechanical integrity program ensures that pipeline system rating of 1.5 times maximum source pressure is maintained. • Combination of materials of construction, internal coating/lining, external coating and cathodic protection assures corrosion risk of SFRM size LOPC is very low. • The MI plan requires a Damage Mechanism Hazard Review to be completed, mitigation plans implemented for all Damage Mechanisms, surveillance plan set to assure mitigation is effective and a PHA is performed if mitigation plan does not meet its objectives.
DEFINITIONS	
Maximum Anticipated Operating Pressure	The maximum pressure that the piping and vessels could be subjected based on process design <u>under normal operating mode</u>. This pressure would normally be below the control system shutdown or PSV set point. In relation to pipeline design, Maximum Anticipated Operating Pressure is to be used as a reference pressure for the Design Factor in a blocked outlet case when there are multiple upstream outlets from the pressure source (e.g., well formation, pump, or compressor).
Maximum Anticipated Source Pressure	The maximum pressure that the piping and vessels could be subjected based on a maximum capacity / shut-in of the pressure source (e.g., well formation, pump, or compressor) in a blocked outlet case. This would represent the well shut in pressure, the centrifugal pump / compressor deadhead pressure, positive displacement pump / compressor discharge cylinder/vessel design pressure or the PSV set point (whichever is highest). In relation to pipeline design, Maximum Source Pressure is to be used as a reference pressure for comparison to the design pressure in a blocked outlet case when the maximum upstream pressure can be determined (i.e. multiple upstream pipeline outlets do not exist).

Table D.4 – SFRM Pipeline Critical Mitigation Elements (CMEs)

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
Pipe: ASME B31.4 / B31.8: Design pressure based on a Design Factor of 0.72 is greater than or equal to the Maximum Anticipated Operating Pressure* *See definition above		0	Overpressure LOPC: Standard design factor per code but DOES NOT provide extra robustness to qualify as a CME		See exact SFRM D&O requirements per line type for information regarding pressure design criteria.
Pipe: ASME B31.4 / B31.8: Design pressure based on a Design Factor of 0.72 is greater than or equal to the Maximum Source Pressure* (e.g., formation, pump, compressor, etc.) *See definition above	SFRM -25	1	Overpressure LOPC: Standard design factor per code and designed to protect against highest source of potential pressure	-All pipeline flanges to be at least Class 150 construction or higher as dictated by the maximum anticipated source pressure -QA/QC measures to verify material -QA/QC measures to assure proper installation -AI/MI measure to detect and monitor degradation to ensure pressure integrity at 1 times maximum source pressure	Intent is to provide an installation that <u>effectively exceeds</u> RAGAGEP for retaining design pressure and providing an <u>acceptable</u> margin of error for defects and degradation. Material selection must be appropriate for service. See exact SFRM D&O requirements per line type for information regarding pressure design criteria.
Pipe: ASME B31.4 / B31.8: Design pressure based on a Design Factor of 0.72 is greater than or equal to the Maximum Source Pressure* (e.g., formation, pump, compressor, etc.), AND the pipe is seamless with 100% RT or UT of	SFRM -26	2	Overpressure LOPC: Standard design factor per code with extra integrity verification measures	-All pipeline flanges to be at least Class 150 construction or higher as dictated by the maximum anticipated source pressure -Materials of construction meet SFRM Design Standards for its service and verified by QA/QC program -Hydro tested per ASME code and based on limiting component (flange or pipe) -MI plan requires a Damage Mechanism Hazard Review to be completed, mitigation plans	Intent is to provide an installation that <u>exceeds</u> RAGAGEP for retaining design pressure and providing an <u>increased</u> margin of error for defects and degradation. Material selection must be appropriate for service.

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
shop and field welded joints *See definition above				implemented for all Damage Mechanisms, surveillance plan set to assure mitigation is effective and a PHA is performed if mitigation plan does not meet its objectives	
Pipe: ASME B31.4 / B31.8: Design pressure based on a Design Factor of 0.5 is greater than the Maximum Anticipated Operating Pressure * *See definition above	SFRM -27	1	Overpressure LOPC: Provides extra margin of error Tier 2/3 ONLY	-All pipeline flanges to be at least Class 150 construction or higher as dictated by the maximum anticipated operating pressure -QA/QC measures to verify material -QA/QC measures to assure proper installation	Intent is to provide an installation that <u>goes beyond</u> RAGAGEP for retaining design pressure and providing a <u>robust</u> margin of error for defects and degradation. Material selection must meet SFRM Design Standard for service. See exact SFRM D&O requirements per line type for information regarding pressure design criteria.
	SFRM -28	1 PR-7	Internal or external corrosion LOPC: Provides extra margin of error. Tier 2/3 ONLY	-Hydro tested per ASME codes and based on limiting component (flange or pipe). -AI/MI measure to detect and monitor degradation and to ensure pressure integrity is maintained to 0.5 design factor rating	
Pipe: ASME B31.4 / B31.8: Design pressure based on a Design Factor of 0.5 is greater than or equal to the Maximum Source Pressure* (e.g., formation, pump, compressor, etc.) *See definition above	SFRM -29	2	Overpressure LOPC: Standard design factor per code and designed to protect against highest source of potential pressure. Tier 2/3 ONLY	-All pipeline flanges to be at least Class 150 construction or higher as dictated by the maximum anticipated source pressure -QA/QC measures to verify material -QA/QC measures to assure proper installation -AI/MI measure to detect and monitor degradation to ensure pressure integrity at 1 times maximum source pressure	Intent is to provide an installation that <u>effectively exceeds</u> RAGAGEP for retaining design pressure and providing an <u>acceptable</u> margin of error for defects and degradation. Material selection must be appropriate for service. See exact SFRM D&O requirements per line type for information regarding pressure design criteria.
	SFRM -30	1 PR-7	Internal or external corrosion LOPC: Provides extra margin of error. Tier 2/3 ONLY		
Pipe: ASME B31.4 / B31.8: Design pressure based on a Design Factor of 0.5 is greater	SFRM -31	3	Overpressure LOPC: Provides enough extra margin of error AND extra quality	-All pipeline flanges to be at least Class 150 construction or higher as dictated by the maximum anticipated source pressure	Intent is to provide an installation that <u>goes significantly beyond</u> RAGAGEP for retaining design pressure

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
than or equal to the Maximum Source Pressure* (e.g., formation, pump, compressor, etc.), AND the pipe is seamless with 100% RT or UT of shop and field welded joints *See definition above			control to allow as three CMEs to protect Multiple and Sensitive receptors. Tier 2/3 ONLY	-Materials of construction meet SFRM Design Standards for its service and verified by QA/QC program -Hydro tested per ASME code and based on limiting component (flange or pipe)	and providing a <u>significant margin of error</u> for defects and degradation. Material selection must be appropriate for service.
	SFRM -32	2 PR-7	Internal or external corrosion LOPC: Provides extra margin of error. Tier 2/3 ONLY	-MI plan requires a Damage Mechanism Hazard Review to be completed, mitigation plans implemented for all Damage Mechanisms, surveillance plan set to assure mitigation is effective and a PHA is performed if mitigation plan does not meet its objectives	
Pipe: ASME B31.4 / B31.8: Design pressure based on a Design Factor of 0.4 is greater than or equal to the Maximum Source Pressure* (e.g., formation, pump, compressor, etc.) *See definition above	SFRM -33	2	Overpressure LOPC: Provides a significant extra margin of error – used when an inherently safer design is warranted.	-All pipeline flanges to be at least Class 150 construction or higher as dictated by the maximum anticipated source pressure -QA/QC measures to verify material -QA/QC measures to assure proper installation -Hydro tested per ASME code and based on limiting component (flange or pipe)	Intent is to provide an installation that is <u>significantly more robust</u> than required by RAGAGEP to retain design pressure and to provide an inherently safer margin of error for defects and degradation. Material selection must be appropriate for service.
	SFRM -34	2 PR-7	Internal or external corrosion LOPC: Provides a significant extra margin of error – used when an inherently safer design is warranted.	-MI plan requires a Damage Mechanism Hazard Review to be completed, mitigation plans implemented for all Damage Mechanisms, surveillance plan set to assure mitigation is effective and a PHA is performed if mitigation plan does not meet its objectives (2 CMEs for corrosion)	
External Pipe Coating	SFRM -35	1 (IF-3)	External Corrosion LOPC: external coating would prevent LOPC	Coating: -QA/QC measures to verify material -QA/QC measures to assure proper installation -AI/MI measure to detect and monitor pipe degradation (ILI, GUL or Hydro tests at increased frequency)	Removed January 2019; External pipe coatings are not credited as CMEs but as a factor in determining the maximum assumed release hole-size.

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
Cathodic Protection (CP)	SFRM -36	1 (PR-22)	External Corrosion LOC: CP would prevent LOC	Cathodic Protection: -QA/QC to verify design -Adequate testing facility -Quarterly checks on system	
PRV on Toxic Hazards Pipeline	SFRM -37	1 (PR-4)	Overpressure LOPC	-PRV must be located where ROE will not reach Fixed Offsite Receptors or arranged in a way that mitigates impact on Fixed Offsite Receptors (e.g., goes to flare) a) If flared, toxic (SO ₂ , H ₂ S, other) ROE modeling and flare design review required. b) If vented to atmosphere, ROE modeling is required to verify that no Fixed Offsite Receptors are impacted above the threshold value. -QA/QC measures to verify design -QA/QC measures to assure proper installation -AI/MI measure to maintain PRV	
Low Probability of SFRM Type Release Due To Internal or External Corrosion in Liquid and Low Pressure Gas Services	SFRM -38	1 (IF-3)	Internal and External Corrosion LOPC. Applicable to: - Lined liquid and low pressure (< 100 psig or 20% of SMYS) gas piping systems - Un-lined seamless liquid and low pressure (< 100 psig or 20% of SMYS) gas piping systems - Stainless steel liquid and low	-QA/QC measures to verify material -QA/QC measures to assure proper installation -Hydro tested per ASME code and based on limiting component (flange or pipe) -AI/MI measure to detect and monitor degradation and to ensure pressure integrity is maintained to 0.5 design factor rating -No unusual situation should be present that would suggest that a leak from the Event Scenario would be more than a small leak	Applicable to limited applications as listed and based on historical analysis that leaks in these services are generally minor that provide sufficient warning to address before larger leaks have time to develop.

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
			pressure (< 100 psig or 20% of SMYS) gas piping		
Clearly Demarcated Pipeline Location Markings and One-Call Program	SFRM -39	1 (PR-19)	Below Ground Mechanical Damage Designed specifically for SFRM Tier 1 application based on existing legal requirements but may require going beyond legal requirement in order to clearly identify where the line is located, direction changes, etc.	-Line routing is properly marked, public awareness and call before excavating program (One-Call) -One Call System in place verified by business unit audit -Oxy-Call system if no other system available (signage with Oxy number to call and rapid Oxy response to mark line) -Measures installed to be included in AI/MI program	
Engineered Passive Excavation Self Announcing Measures (dye, ribbons, encasement, etc.)	SFRM -40	1 (PR-19)	Below Ground Mechanical Damage	-QA/QC to assure design and installation will meet the intent of this CME	Measure would make it obvious to excavator operator before reaching the pipe depth that a pipeline is below
Robust Engineered Pipeline Markings and One-Call Program	SFRM -41	1 (PR-19)	Below Ground Mechanical Damage Designed specifically for SFRM Tier 2 and 3 applications beyond existing legal requirements.	Examples include a) Close interval robust signs (substantial posts with large print, weather resistant signs) following centerline of pipeline assuring that any change in direction is obvious b) Same as previous example with the addition of cable laid on surface between posts or other physical barrier for the length of pipeline that could have an ROE that reaches receptors. -One Call System in place verified by business unit audit	Measures would so dominate that digging would be a willful decision by excavator operator to disregard the warning

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
				-Oxy-Call system if no other system available (signage with Oxy number to call and rapid Oxy response to mark line) -Measures installed to be included in AI/MI program	
Pipeline Located Under Paved Street in an Urban Environment and Proven One-Call Program	SFRM -42	1 (PR-19)	Below Ground Mechanical Damage	-BU has experience with responding One-Call agency and has evidence that One-Call is rigorously enforced	One Call System in place verified by internal self-audit Measures would be so dominant that digging would be a willful decision by excavator operator to disregard the warnings and One-Call requirements
SFRM Vehicle Impact Barriers	SFRM -43	2 (PR-12)	Vehicle Impacts to above ground piping having elevated risk: Barriers engineered for this purpose would have a very high probability of preventing damage that would cause LOPC	-Barriers designed to OOG Engineering Standard -Barriers are required for instances where there is an identified elevated probability that the piping could be impacted resulting in such a way as to cause LOPC	
Stainless Steel WAG Piping	SFRM -44	1 (IF-3)	Internal Corrosion LOPC	-QA/QC measures to verify material -QA/QC measures to assure proper installation -AI/MI measure to detect and monitor pipe degradation (ILI, GUL or Hydro tests at increased frequency)	Internal pipe coating CME removed January 2019; internal pipe coatings are not credited as CMEs but as a factor in determining the maximum size of release.
High Pressure/Flow/Level Device Interlocked to Shutdown Function	SFRM -45	1 (PR-6)	Overpressure, Internal or External Corrosion LOPC – see comments for specific cases	-Requires interlocked shutdown to prevent or mitigate scenario related to pressure/flow/level -QA/QC measures to assure proper installation and ongoing reliability of the interlock system	No use of SCADA or other radio frequency or a polling operation. A polling operation could require a significant amount of time before

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
					shutdown and use of SCADA or radio frequency is not reliable enough for a CME. Examples include: CO₂ & H₂O flood flow lines: High Flow line pressure alarm and shutdown of artificial lift motor, or close flowing well SDV CO₂ & H₂O Prod Inlet Header: High - High inlet header pressure alarm and shutdown interlocked to close satellite production separator inlet ESD/SDV Prod Trunk lines: High pressure shutdown device on Production Truck Line interlocked to trip the Satellite Production Separator inlet ESD/SDV and associated wells' artificial lift motor power or close flowing well ESD/SDV <i>(a KME for the shutdown of associated wells)</i> Prod Trunk lines: High pressure shutdown device on transfer line interlocked to close the Satellite Production Separator outlet SDV (ensure that transfer pump

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
					shutdowns also if so equipped) 5. Gas Gathering Satellite: High Satellite Production Separator pressure alarm and high-high shutdown interlocked to associated well artificial lift motor trip or flowing well shutdown valve and production separator shutdown valve (<i>a KME for the shutdown of associated wells</i>) 6. Gas Gathering CTB: CTB LP Compressor high-high discharge pressure shutdown of the LP Compressor and close Compressor inlet SDV 7. H2O Inj Trunk lines: Water Injection Pump(s) high-high discharge pressure shutdown of the Water Injection Pump(s) (source of pressure) 8. CO2 Pipelines >100 PPM H2S: High - High CO2 Pipeline discharge pressure device interlocked to shutdown RCF / Plant Compressors 9. CO2 & WAG Injectors: High Injection Pressure Shutdown at the WAG skid or header

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
					interlocked to the choke/ESD/SDV Additional high pressure/flow/level devices interlocked to shutdown function may be credited as a CME but not considered as a D&O requirements. 1. CO2 & H2O Prod Inlet Header: ESD/SDV at Satellite inlet interlocked to shutdown associated wells' artificial lift motor power or close flowing well ESD/SDV with automated call out to Production Tech
Low Pressure/Flow/Level Device Interlocked to Shutdown Function	SFRM -46	1 (PR-6)	Overpressure, Internal or External Corrosion LOPC – see comments for specific cases	-Requires interlocked shutdown to prevent or mitigate scenario related to pressure/flow/level -QA/QC measures to assure proper installation and ongoing reliability of the interlock system	No use of SCADA or other radio frequency or a polling operation. A polling operation could require a significant amount of time before shutdown and use of SCADA or radio frequency is not reliable enough for a CME. Examples include: 1. CO2 & H2O flood flow lines: Low Flow line pressure alarm and shutdown of artificial lift motor, or close flowing well SDV 2. CO2 & H2O Prod Inlet Header: Low - Low inlet header pressure alarm and

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
					shutdown interlocked to satellite production separator inlet ESD/SDV 3. Prod Trunk lines: Low level shutdown device on the Satellite Inlet Separator with a low-low shutdown of Satellite Inlet ESD/SDV interlocked to associated wells artificial lift motor power or flowing well ESD/SDV (<i>a KME for the shutdown of associated wells</i>) 4. Gas Gathering Satellite: Low pressure device downstream of the Satellite Production Separator pressure control valve interlocked to the Satellite inlet SDV 5. Gas Gathering CTB: CTB LP Compressor low-low discharge pressure shutdown of the LP Compressor (source of pressure) 6. Gas Gathering CTB: Redundant Low pressure device on the discharge of the CTB LP Compressor at the CTB Battery limit interlocked the CTB Separator inlet SDV (applicable to non-looped

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
					Gas Gathering Systems only) 7. H2O Inj Trunk lines: Water Injection Pump(s) low-low discharge pressure shutdown of the Water Injection Pump(s) 8. CO2 Pipelines >100 PPM H2S: Low - Low CO2 Pipeline pressure device at the RCF/Plant tailgate interlocked to shutdown an ESD/SDV at the outlet of the RCF/Plant on trunk line(s) 9. CO2 & WAG Injectors: Low Injection Pressure Shutdown device at the WAG skid or header interlocked to the choke/ESD/SDV Additional low pressure/flow/level devices interlocked to shutdown function may be credited as a CME but not considered as a D&O requirements.
Production Flowline ESD/SDV Closure tied to High/Low Pressure	SFRM -47	1 (PR-6)	Overpressure, Internal or external corrosion LOPC – see comments for specific cases	-Requires interlocked shutdown to prevent or mitigate scenario related to pressure/flow/level -QA/QC measures to assure proper installation and ongoing reliability of the interlock system	No use of SCADA or other radio frequency or a polling operation. A polling operation could require a significant amount of time before shutdown and use of SCADA or radio frequency is not reliable enough for a CME.

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
					Examples include: ESP flow lines: Additional pressure device on the discharge of the ESP upstream of the ESD/SDV flow line valve wired/instrumented independently set to trip Choke/ESD/SDV Flowline valve on high or low flow line pressure Flowing Well Flow lines: Redundant ESD/SDV wired/instrumented independently of the flow line choke set to trip on high or low flow line pressure Beam Unit Flow lines: ESD/SDV wired / instrumented independently of the Pump Off Controller high and low pressure trip on the flow line set to trip on high or low flow line pressure
Beam Unit Stuffing Box Failure Applied As a Pressure Relief Device for the Production Flow Line	SFRM -48	1 (PR-4)	Overpressure to Flow Line caused by pressure from Beam Unit (e.g. under a blocked outlet case)	-Requires: - documentation that stuffing box will fail below the pressure components' rating by a minimum of 500psig 0.5-inch ROE from stuffing box failure will not impact public receptors at the SFRM thresholds (0.5-inch assumed to be equivalent hole size for the stuffing box packing failure)	ROE is Radius Of Exposure corresponding to the SFRM screening thresholds of OOG 60.400.302 HES Risk Screening.

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
				-QA/QC measures to assure proper installation and ongoing reliability of stuffing box's failure pressure (e.g., MOC of design/equipment changes) If applied, confirm flow line design and integrity to ensure stuffing box would activate at lowest pressure.	
Pipeline Differential Pressure Device Interlocked to a Shutdown Function	SFRM -49	1 (PR-6)	Overpressure, Internal or external corrosion LOPC – see comments	-Requires interlocked shutdown to prevent or mitigate scenario related to pressure/flow/level -QA/QC measures to assure proper installation and ongoing reliability of interlock system	
Car Sealed or Locked with Car Seal Program	SFRM -50	1 (IF-7)	Overpressure LOPC	-Manual valve could be locked open or car sealed open -Must be supported by a Car Seal Program.	
Fiber Optic Leak Detection Tied to a Shutdown Device	SFRM -51	1 (PR-6)	Overpressure, Internal or external corrosion LOPC	-Requires interlocked shutdown to prevent or mitigate scenario related to pressure/flow/level -QA/QC measures to assure proper installation and ongoing reliability of the interlock system and leak detection	Due to reliability concerns based on current technology, this design/CME option should be used in very limited applications based on current technology and should be vetted and approved by the Core Risk Group prior to use.
Pipeline Inventory Isolation Valve Stations	SFRM -52	1	Internal or external corrosion LOPC, Below Ground Mechanical Damage, Vehicle Impacts to above ground piping Applicable to specific pipelines with SFRM exposure where	Inventory isolation valve stations limiting the quantity released on either of the following: -Gas gathering lines -CO ₂ distribution and supply lines -Other lines with potential SFRM exposure where inventory reduction would prevent SFRM exposure -QA/QC measures to assure proper installation and ongoing reliability	

SFRM PIPELINES CRITICAL MITIGATION ELEMENTS (CMEs)					
CME	SFRM CME	# CME	Applicability	KME Requirements	Comments
			inventory reduction would prevent SFRM exposure		

Appendix E – Drilling and Well Servicing Risk Assessment

1. **Overview:** Drilling and well servicing risks do not lend themselves to the standard OOG CME methodology and require conventions for deriving risk levels that are comparable to other HSE risks. The standard HSE Consequence Levels will be applied; however, OOG Drilling, Well Servicing, and HSE Risk Management have jointly developed unique mitigations elements, Critical Mitigation Systems (CMS), for evaluating the Probabilities associated with these activities.
2. **Risk Register Categories:** The following eight (8) Event Scenarios are used to evaluate these risks (each will be a separate Risk Register):
 - a. Drilling
 - i. Drilling Loss of Well Control – Overbalanced Drilling
 - ii. Drilling Loss of Well Control – Underbalanced Drilling
 - iii. Drilling Rig Operations
 - iv. Drilling Rig Mobilization and Demobilization (Land-Based or Offshore Rigs)
 - b. Well Servicing (Interventions & Maintenance)
 - i. Well Servicing Loss of Well Control – Overbalanced
 - ii. Well Servicing Loss of Well Control – Underbalanced
 - iii. Well Servicing – Higher Risk Activities
 - iv. Well Servicing Mobilization and Demobilization (Land-Based or Offshore Rigs)
3. **Implementation:**
 - a. Each Business Unit or independent drilling project (as they apply) must complete a single Risk Register for each of the Event Scenarios listed above based on their overall drilling or well servicing activities. The risks must be re-assessed annually to ensure that mitigations are implemented effectively. When the MOC program identifies changes that impact the risk level, existing risk registers must be revised and notifications made in accordance with Section 3.4 of this procedure.
 - b. The Drilling Risk Registers must be developed as a joint effort between the BU drilling and BU HSE functions, and the Well Servicing Risk Registers jointly by the BU well servicing and BU HSE functions. The Well Servicing – High Risk Activities will be developed by either the BU drilling or well servicing group, as applicable to the organization of each BU, jointly working with the HSE functions. Drilling and Well Servicing will continue to comply with the OOG and BU Drilling and Well Servicing PSPGs (e.g. well design criteria, MOCs, etc.) in addition to the HSE risk register development process.
 - c. Drilling and Well Servicing Event Scenarios must be evaluated using unique Critical Mitigation Systems (CMSs) as defined in the Event Scenario Tables (Section 6) for determining Probabilities. The estimated Consequence Level is to be coupled with the Probability Level derived from the number of CMSs credited to establish the associated risk level.
 - d. For Event Scenario 7, Well Servicing High Risk Activities, BU's must assess the risk of each applicable discrete high-risk activity. The Risk Register submitted must reflect information

for the discrete activity with the highest risk level among all activities evaluated. If multiple activities carry the same highest risk level, then the Register should collectively list these activities in the comments section.

4. **Critical Mitigation System (CMS):** A group of mitigation measures that in total will provide a robust barrier to prevent an undesired non-process Event Scenario. Failure of one CMS shall not prevent another CMS from performing its function.

A CMS is a layer of protection for non-process Event Scenarios, equivalent to a Critical Mitigation Element (CME) for a process Event Scenario, that may be credited to reduce the probability of the Event Scenario one level on the Risk Matrix (e.g., from 1/10 yrs. to 1/100 yrs.) To credit a CMS, the associated mitigation elements must be validated as in compliance with the applicable standards and requirements. Unless otherwise noted, an OOG approved variance is to be obtained if the standards and requirements cannot be implemented.

5. **Characterization:** Each BU must provide the following statistical information in Risk Registers:

Well Control	(4 Risk Registers): Estimated annual number of wells drilled or serviced each year and the estimated annual number of wells drilled or serviced that have an offsite E Level Risk.
Rig Operations	(1 Risk Register): Estimated annual number of wells drilled each year, the number of well drilled by automated rigs each year and the estimated annual number that have an offsite E Level Risk.
Well Servicing – Higher Risk Activities	(1 Risk Register): Estimated annual total number of well servicing operations that occur each year, the estimated annual number of operations each year that the Risk Register represents and the estimated annual number that have an offsite E Level Risk.
Rig Mobilization / Demobilization	(2 Risk Registers): Estimated annual number of moves each year (count each mob and demob separately except count moves from one Oxy well to another as one move) for land based rigs and offshore rigs, the number of moves that are modular rigs and the estimated annual number that have an offsite E Level Risk.

6. **Event Scenario Tables:** Each of the following tables represents a separate risk Event Scenario. Within each table are instructions for developing the Event Scenario Consequences and a list of CMSs that are available to reduce the Probability of each Event Scenario's Consequences.

Event Scenario 1: Drilling Loss of Well Control – Overbalanced Drilling - (2 CMSs Available)

- Consequences Level based on site specific circumstances (public receptors, environmental receptors and number of people on site – see Appendix A – OOG HSE Risk Matrix, above)

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Well Design (D-1) 1 CMS	Casing Design (casing integrity)	60.605.040 - Tubular Design	Implemented and in compliance as applicable or approved OOG variance in effect
	BOPE	60.604.700 - Well Control	
	Standards	60.600.400 - Management of Change 60.603.710 - Wellbore Anti-Collision 60.605.400 - Wellhead & Tree Design	
	Contractor Competency	Supervisors with IADC/IWCF Well Control Certification (Completion & Workover, Coiled Tubing, Wireline and Snubbing), Crews with well control awareness training.	
Overbalance (D-2) 1 CMS	Fluid Design	Pore Pressure/Frac Gradient Analysis as supported by 60.601.830 - Well Construction Process 60.603.900 - Drilling Fluids	Implemented and in compliance as applicable or approved OOG variance in effect
		Leak-Off Test/Formation Integrity Test and Kick Detection as supported by 60.604.700 - Well Control	
	Standards	60.600.400 - Management of Change	
	Contractor Competency	Supervisors with IADC/IWCF Well Control Certification (Completion & Workover, Coiled Tubing, Wireline and Snubbing), Crews with well control awareness training.	

Event Scenario 2: Drilling Loss of Well Control – Underbalanced Drilling - (2 CMSs Available)

- Consequence Level based on site specific circumstances (public receptors, environmental receptors and number of people on site – see Appendix A – OOG HSE Risk Matrix, above)

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Well Design (D-1) 1 CMS	Casing Design (casing integrity)	Std 039 60.605.040 - Tubular Design	Implemented and in compliance as applicable or approved OOG variance in effect
	BOPE	Std 033 60.604.700 - Well Control	
	Standards	60.600.400 - Management of Change 60.603.710 - Wellbore Anti-Collision 60.605.400 - Wellhead & Tree Design	
	Contractor Competency	Supervisors with IADC/IWCF Well Control Certification (Completion & Workover, Coiled Tubing, Wireline and Snubbing), Crews with well control awareness training.	
Primary Well Control (D-7) 1 CMS	Underbalanced Drilling Package / Managed Pressure Drilling Package	60.604.700 - Well Control This package represents a separate set of equipment (rotating head, choke manifold, gas buster) that provides a pressure-retention surface seal. To be claimed as a CMS, the control system including related equipment and devices must also be independent from the BOPE setup.	Implemented and in compliance as applicable or approved OOG variance in effect

Event Scenario 3: Drilling Rig Operations - (2 CMSs Available)

- Consequence Level typically Level 4 based on 2-3 people on the drill floor.
 - o NOTE: The Consequence Level for an offshore installation may be higher due to potentially larger number of barge personnel working on a well, in addition to higher production loss from adjacent well(s) if impact occurs.

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
High Standards for Rig, Crew and Supervision (D-3) 1 CMS	Contractors Competency	Short Service Employee (SSE), Training, Turnover, Supervision (RM)	Implemented and in compliance as applicable
	HESMS PSPG's & BU Performance Standards	JSA's, BBS, PPE, Work Permits, Hazard ID; Bridging documents in place of BU PSPGs, if appropriate OPSP F-1 Fall Protection	
	Rig Equipment	Conformance with applicable API specifications and recommended practices for Rig Acceptance Criteria (Magnetic Integrity Test (MIT)/ Magnetic Particle Inspection Criteria), Automation, Contracts Requirements	
	Operating PSPG's	Drilling & SimOps Procedures (Oxy & Contractor)	Implemented and in compliance as applicable or approved OOG variance in effect
Automated Rig (D-4) 1 CMS	Personnel Not Present	Personnel not normally present on rig floor and pipe handing areas during normal drilling operations <i>BU must obtain OOG Drilling concurrence to apply this CMS.</i>	Implemented and in compliance as applicable

Event Scenario 4-1: Rig Mobilization and Demobilization, Drilling (Land-Based Rigs, 2 CMSs Available)

- Consequence Level typically Level 3 (1 PII) or 4 (2-3 PIIs) based on local conditions.

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Robust Rig Movement Planning and Supervision (D-5) 1 CMS	Written Rig Move Plan	Rig move will not occur if hazards increase due to: - Weather (sustained winds above 40 mph or gust above 60 mph; blinding - snow/rain/dust/sand; sleet; lightning; severe road or equipment icing, severely muddy roads; tornado/hurricane/flood warnings) - Terrain (inadequate road/route preparation or road condition due to grade) - Time limits (if distance is too far, prepare & plan staging areas for overnight staging; 16 hrs/day limitation)	Implemented and in compliance as applicable or approved OOG variance in effect
		Moves are sequenced to minimize site congestion, multiple lifts, etc.	
		Site schematics with equipment layout are utilized.	
		Cranes are certified for pick weights, lifting & have qualified operators, designated banksman & riggers. Lifting devices have current certification & appropriate for lift.	
		Designated spotters with appropriate training and communications capabilities to truck drivers, equipment operators & other directly involved parties.	
		Route on Oxy property are driven/walked to identify any potential hazards	
		Written rig move plan provided by contractor and Oxy rig move guideline followed by contractor (60.608.440)	
	Controls	Rig Manager on location monitoring rig moves	Implemented and in compliance as applicable
		Drill Site Manager on location monitoring rig moves	
		HSE Representative on location monitoring rig moves	
		Rig trucking supervisor on location monitoring rig moves	
Rig Design (D-6) 1 CMS	Modular Rigs	1) Individual components have been configured to reduce significantly the amount of disassembly of such components (e.g., top drive, traveling block arrangement that stay intact during a move). 2) Other systems that are similarly skid or trailer mounted includes substructure, generator/air compressor, fluid pumps, mud tanks/solid controls and BOP equipment. 3) Maximum average of 40-60 loads	Implemented and in compliance as applicable

Event Scenario 4-2: Rig Mobilization and Demobilization, Drilling (Offshore Rigs, 3 CMSs Available)

Note: Mobilization-De-mobilization only includes the rig movement & pre-load not rig-up/rig down (RU/RD) which is covered under the Rig Operations register. Rig Move refers to mobilization and demobilization collectively.

- **Consequence Level typically Level 5 (> 3 PIs) based on ship collision or punch through with impact to platform leading to loss of containment (possible fire, explosion, toxic event); PD-LOR Consequence Level 4 or 5 based on location**

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Planning & Personnel (D-8) 1 CMS	Rig Move Procedure	Location approval by Marine Warranty Surveyor (MWS) (position, water depth, expected penetration (historical data footprints, soil sample for new location), stand-off distance)	Implemented and in compliance as applicable
		Rig approach plan	
		Weather forecast (2 independent sources)	
		Seabed survey – 500 x 650 ft. cleared area defined – pipelines marked where necessary – debris cleared; 6 weeks maximum validity	Implemented and in compliance as applicable or approved OOG variance in effect
		Equipment depressurization requirements	
		Weather criteria for move – rig move not allowed if wind speed above 15 knots or sea state 4 feet or higher	
		Tow vessel requirements & suitability	
	Rig Move Planning Meeting & Pre-Move Checks	Pre-rig move audit and survey of intended platform	Implemented and in compliance as applicable
		Tow vessel captains briefed by Rig Mover (RM) – duties assigned	
		Rig Crew undertake pre-move checks as per Rig move procedure	
		Watertight Integrity Confirmed - Deck cargo secured - readiness to move confirmed by RM, MWS, Barge Engineer (BE)	
		Variable deck load calculated and within operating parameters	
		Rig move briefing with senior personnel on rig	
		Contractor Emergency Response Plan approved by Oxy	
	Minimum Qualified Personnel	Licensed Rig Mover Independent Licensed Marine Warranty Surveyor Licensed Vessel Crews Rig minimum manning level as per Flag State Requirements (e.g. ABS) Oxy Rep Onboard Licensed OIM Onboard Rig Crew with Emergency Response Training	

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Rig Move Controls (D-9) 1 CMS	Isolation of Wells and Wellhead Jackets	Wells are isolated with two proven barriers	Implemented and in compliance as applicable or approved OOG variance in effect
		Subsea hydrocarbon lines isolated	
		Topsides hydrocarbon lines depressurized to 150 psig maximum	
		Rig Mover dedicated channel to communicate with vessels and jacking control room	
		Move-on/Move-off undertaken in daylight hours only	Implemented and in compliance as applicable
		Single point control (Rig Mover) on transom for final positioning	
		Soft pinning in surveyed zone (rig box)	
		Rig makes final approach onto Jacket with Spud Cans dragging on seabed	
		Rig confirmed in final acceptable position	
		Rig commences preload operation at 5 foot air gap as per MWS procedure	
		During preload operations - tow vessel in attendance & no helicopter or crane operations allowed	
		No personnel or vessels allowed alongside jacket until preload is dumped	
		ESD function tested and handed to rig. ESD manned during jacking operations	
		PM and 3rd party inspection on jacking components - redundancy built into jacking system	
		Leg penetration checks & Spud Can Survey carried out via remote operated vehicle	
Towing & Support Equipment (D-10) 1 CMS	Certified rig move equipment	3 x Anchor Handling Tug Supply Vessels with sufficient bollard pull - approved, assigned and accepted by RM & MWS	Implemented and in compliance as applicable
		Rig tow equipment inspected as part of 6-monthly 3rd party NDT program	
		Emergency response & evacuation equipment available	

Event Scenario 5: Well Servicing Loss of Well Control – Overbalanced (2 CMSs Available)

- Consequence Level based on site specific circumstances (public receptors, environmental receptors and number of people on site – see Appendix A – OOG HSE Risk Matrix, above)

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Well Control (W-1) 1 CMS	BOPE	Well Control Standard meets or exceeds API STD 53 & RP 59	Implemented and in compliance as applicable or approved OOG variance in effect
	Standards	60.630.001 STD Well Control for Workover and Well Servicing Rig Operations (as to BOPE) 60.630.010 STD Well Control for Coiled Tubing, Snubbing, and Wire-Line Operations	
	Contractor Competency	Supervisors with IADC/IWCF Well Control Certification (Workover, Coiled Tubing, Wireline and Snubbing), Crews with well control awareness training.	
Overbalance (W-2) 1 CMS	Fluid Design	Exceeds Formation Pressure	Implemented and in compliance as applicable or approved OOG variance in effect
		Well Control Standard meets or exceeds API STD 53 & RP 59	
		60.630.001 STD Well Control for Workover and Well Servicing Rig Operations (as to controlling the Maximum Anticipated Surface Pressure)	
	Contractor Competency	Supervisors with IADC/IWCF Well Control Certification (Completion & Workover, Coiled Tubing, Wireline and Snubbing), Crews with well control awareness training.	

Event Scenario 6: Well Servicing Loss of Well Control – Underbalanced (2 CMSs Available)

Underbalanced well servicing operations, whether intentional or situational, may include wireline/slickline, coiled tubing, and workover rig work, the latter of which could include snubbing.

- **Consequences Level: based on site specific circumstances (public receptors, environmental receptors and number of people on site – see Appendix A – OOG HSE Risk Matrix, above)**

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Well Control (W-1) 1 CMS	BOPE	Well Control Standard meets or exceeds API RP 53 & 59	Implemented and in compliance as applicable or approved OOG variance in effect
	Standards	60.630.001 STD Well Control for Workover and Well Servicing Rig Operations (as to BOPE) 60.630.010 STD Well Control for Coiled Tubing, Snubbing, and Wire-Line Operations	
	Contractor Competency	Supervisors with IADC Well Control Certification (Workover, Coiled Tubing, Wireline and Snubbing), Crews with well control awareness training.	
Primary Well Control (W-6) 1 CMS	Based on the underbalanced intervention operation, the primary well control equipment is listed as follows: Wire-Line: Stuffing Box (slick-line) / Grease Injector System (braided or electrical line) Coil Tubing: Stripper Assembly Snubbing: Stripping BOP / Stripping Element	60.630.010 STD Standard for Well Control for Coiled Tubing, Snubbing, and Wire-Line Operations. This CMS represents a separate set of equipment that provides a pressure-retention surface seal. To be claimed as a CMS, the associated control system, including related equipment and devices, must also be independent from the BOPE setup.	Implemented and in compliance as applicable or approved OOG variance in effect

Event Scenario 7: Well Servicing – Higher Risk Activities

High Risk Activities Involving Work Over, Well Intervention, Well Clean-Up & Well Testing (including Stimulation & Completions). Note: Risks associated with Loss of Well Control are not included in this Risk Register.

Regardless of BU functional group (drilling, well servicing) involved in performing the service, this category specifically addresses high risk activities such as pressure pumping, wireline/slickline/coiled tubing operations, rig lifting, rig up/rig down, high pressure testing, fishing & jarring, plug abandonment, and other high risk activities related to completions, well stimulation, well cleanup and well testing as listed in the summary table below.

The BU is to conduct a risk assessment of each general high-risk activity listed in Table E.7 to establish the related Consequence (based on number of personnel involved) and the related Probability (based on implementation of available CMSs). This assessment should be maintained on file by the BU, and updated as the Consequence or Probability levels change. The BU must update the Register for Event Scenario 7 at least annually as part of the BU Risk Inventory process, together with other drilling and well servicing Registers.

One Risk Register is to be submitted for Event Scenario 7 as a whole. The Risk Register submitted must reflect the information associated with a discrete activity listed in the table that has the highest risk level among all activities. If multiple activities carry the same highest risk level, then the Register information can be based on any one of the highest risk discrete activities listed in Table E.7. List the title(s) for all other highest risk activities in the Register comments section.

The specific high-risk activity that are covered by this Register are listed in the table below together with available CMSs in each case. For the purposes of this evaluation, Well Servicing has been classed into four main categories: (1) Initial Completion, (2) Intervention, (3) Workover, and (4) Well Cleanup/Testing Operations. Each category carries its unique set of high risks that constitute a means for grouping related hazards and available CMSs.

Initial Completion: An operation carried out for an oil and gas well at the beginning of its life that enables initiating production from the well, e.g. via perforating and/or stimulation. The initial completion typically involves adding hardware. Examples of well completion operations include perforating (wireline or tubing-conveyed) and setting plugs, stimulation (high pressure pumping, including multi-stage fracture stimulation), slick or braided line ops, coil tubing, and rig work (including downhole gauge and artificial lift installation), possibly including snubbing.

Intervention: An operation carried out for an oil and gas well during or at the end of its life that alters the state of the completion, provides well diagnostics, or manages the production of the well e.g. via stimulation. Generally, intervention may not involve adding or removing any hardware (completion) to the existing wellbore, and during the activity, the existing completion equipment typically remains in place. Examples of well intervention operations include coiled tubing, high pressure pumping, wireline, slickline, braided line.

Workover: An operation that involves pulling and replacing a production equipment, and possibly recompleting the well. The workover may involve killing the well or conducting snubbing operations, releasing and/or setting a production packer or tubing anchor, removing the wellhead and/or Xmas tree assembly, including the flow line, installing a blow-out preventer, lifting the tubing hanger from the casing head, and pulling the production equipment. The well would then be maintained and/or rehabilitated and returned to its original configuration, or re-completed to a new zone, likely with a new configuration; in any case a new, redesigned, or alternative artificial lift system may be installed.

Well Cleanup & Testing Operations: Cleanup encompasses flowback of well fluids to a frac tank, open top tank, or pit, typically after the initial completion or a recompletion workover, possibly after an intervention. This setup may include flow of separated gas to a flare stack or a pit. Well testing encompasses flowback to a well testing (measurement) skid for separation of gas and liquids, and directing separated gas to a flare.

Table E.7: Well Servicing Higher Risk Activities

High Risk Activity	Applicable to Well Servicing Category	Description of Activity	Primary Hazard(s)	Potential Consequences	Available CMS(s)
7.1 Rig Related Lifting	Initial Completion Intervention Workover Well Clean-up & Well Testing Operations	Lifting of tubulars & ancillary equipment (e.g. using cranes & forklifts)	Dropped load / objects	Personnel: 2-3 PIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-7 Materials Handling, Rigging & Lifting, and Dropped Objects Controls (1 credit)
7.2 Rig Up / Rig Down	Initial Completion Intervention Workover Well Clean-up & Well Testing Operations	Typical equipment to be rigged up/down includes hoist, coil tubing unit, wireline unit, flowback/well test package etc.	Dropped objects	Personnel: 2-3 PIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-7 Materials Handling, Rigging & Lifting, and Dropped Objects Controls (1 credit)
			Caught between (as part of offloading equipment from truck, or other staging equipment)	Personnel: 1 PII	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-7 Materials Handling, Rigging & Lifting, and Dropped Objects Controls (1 credit)
7.3 Continuous Rod Removal and Re-Insertion	Workover	This operation entails rod removal and re-insertion for beam pump or positive cavitation pumps using equipment suspending over wellhead.	Dropped objects	Personnel: 1-2 PIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-7 Materials Handling, Rigging & Lifting, and Dropped Objects Controls (1 credit)
7.4 Beam Pump Horse Head Removal / Installation	Workover	This operation entails removal of beam unit horse head using a mobile crane while working at height with tools.	Dropped objects	Personnel: 1-2 PIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-7 Materials Handling, Rigging & Lifting, and Dropped Objects Controls (1 credit)
	Workover		Un-isolated Energy	Personnel: 1-2 PIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) 1 Additional probability reduction credit (if installed): Secondary means of locking unit from rotation

High Risk Activity	Applicable to Well Servicing Category	Description of Activity	Primary Hazard(s)	Potential Consequences	Available CMS(s)
7.5 Sand Cleanout Operations - via High Pressure Pumping	Initial Completion	This operation entails using a high pressure pump to deliver fluid into the well to circulate sand/fluid back to surface. The operation is typically performed using a hoist or a coil tubing unit.	Dropped objects (if a hoist is employed)	Personnel: 1-2 PIIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-7 Materials Handling, Rigging & Lifting, and Dropped Objects Controls (1 credit)
	Workover		Overpressure and failure of iron or hoses with impact hazard	Personnel: 1-2 PIIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-8 Control of High Pressure Pumping/Testing Operations
7.6 Perforating Gun Lifting, Insertion, & Removal	Initial Completion	This operation entails movement of tool sets that are used to perforate production casing using a series of explosive charges. The operation is typically performed using a wireline unit. Tubing conveyed perforating guns employ a hoist.	Dropped objects (about 700 to 800 pounds)	Personnel: 2-3 PIIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-7 Materials Handling, Rigging & Lifting, and Dropped Objects Controls (1 credit)
	Workover (Tubing Conveyed Perforating Guns)		Accidental detonation of explosive charge at surface	Personnel: 2-3 PIIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-9 Explosives Handling (1 credit)
7.7 High Pressure Testing	Initial Completion Intervention Workover Well Clean-up & Well Testing Operations	This operation entails pressure testing of BOPs, wellhead equipment, casing and tubing strings, and flow lines using a high pressure pumping unit.	Overpressure and failure of iron or hoses with impact hazard	Personnel: 1-2 PIIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-8 Control of High Pressure Pumping/Testing Operations (1 credit)
7.8 Well Acid Stimulation - via High Pressure Pumping	Initial Completion Intervention Workover	This operation entails pumping acid into the reservoir for well stimulation using a high pressure pumping unit.	Overpressure and failure of iron or hoses with impact hazard	Personnel: 1-2 PIIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-8 Control of High Pressure Pumping/Testing Operations

High Risk Activity	Applicable to Well Servicing Category	Description of Activity	Primary Hazard(s)	Potential Consequences	Available CMS(s)
7.9 Fracking with Proppants) - via High Pressure Pumping	Initial Completion	This operation entails pumping proppants into the reservoir to increase well productivity using a high pressure pumping unit.	Overpressure and failure of iron or hoses with impact hazard	Personnel: 1-2 PIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit)
	Intervention			Property Damage: Potential Frac Fleet Damage based on value of equipment	W-8 Control of High Pressure Pumping/Testing Operations
	Workover				
7.10 Nitrogen Lifting - via High Pressure Pumping	Initial Completions	This operation entails using a high pressure pumping unit usually with a coiled tubing unit to inject liquid nitrogen into a stagnated well to lighten the fluid column and induce natural flow.	Overpressure and failure of iron or hoses with impact hazard	Personnel: 1-2 PIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit)
	Intervention				W-8 Control of High Pressure Pumping/Testing Operations
	Workover		Exposure to low temperature liquid nitrogen	Personnel: 1 PII	W-3 - Robust Management of Well Servicing Operational Risks (1 credit)
					W-8 Control of High Pressure Pumping/Testing Operations
7.11 Fishing & Jarring	Initial Completion	Fishing is performed to retrieve lost items downhole using a wireline unit, coil tubing unit or hoist. A jar is a tool employed to remove stuck tools or pipe downhole during fishing operations. Due to inherent tension used to un-lodge the stuck tools/pipe, jarring has the potential for significant recoil effect on the overhead rig components. Jarring can be performed using a wireline unit, coil tubing unit or hoist.	Dropped object hazard due to jarring action	Personnel: 2-3 PIIs due to impact	W-3 - Robust Management of Well Servicing Operational Risks (1 credit)
	Intervention				W-7 Materials Handling, Rigging & Lifting, and Dropped Objects Controls (1 credit)
	Workover				
	Initial Completion		Exceed the load rating of the rig structure due to over-pull leading to structural failure (e.g. derrick, block, drill line, elevators etc.)	Personnel: 2-3 PIIs due to impact	W-3 - Robust Management of Well Servicing Operational Risks (1 credit)
	Workover				W-7 Materials Handling, Rigging & Lifting, and Dropped Objects Controls (1 credit)
7.12 Well Testing Operations	Initial Completion	This encompasses flowback to a well testing (measurement) skid for separation of gas and liquids, and directing separated gas to a flare.	Toxic/flammable gas release	Personnel: 2-3 PIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit)
	Intervention				W-10 Control of Flowback, Well Clean-up & Well Testing Operations (1 Credit)
	Workover				

High Risk Activity	Applicable to Well Servicing Category	Description of Activity	Primary Hazard(s)	Potential Consequences	Available CMS(s)
7.13 Well Clean-up Operations	Initial Completion	This encompasses flowback of well fluids to a frac tank, open top tank, or pit. This setup may include flow of separated gas to a flare stack or a pit.	Toxic/flammable gas release	Personnel: 2-3 PIIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-10 Control of Flowback, Well Clean-up & Well Testing Operations (1 credit)
	Intervention		Internal explosion within the receiving equipment due to flammable atmosphere	Personnel: 2-3 PIIIs Property Damage: Potential asset damage based on equipment value	W-3 - Robust Management of Well Servicing Operational Risks (1 credit) W-10 Control of Flowback, Well Clean-up & Well Testing Operations (1 credit)
7.14 Logging with Radioactive Source	Initial Completion	This operation entails using a wireline unit to lower the logging tools with radioactive source into the well formation. The purpose is to evaluate the reservoir and map the formation.	Exposure to radioactive material	Personnel: 2-3 PIIIs	W-3 - Robust Management of Well Servicing Operational Risks (1 credit)
	Intervention			Environmental Impairment: Based on potential penalties and remediation required.	W-11 Radioactive Source Controls (1 Credit)
7.15 Snubbing Operations	Initial Completion	Typically performed on a live well, this operating involves conveying tubing/pipe into well under pressure most commonly for well control, complex fishing operations, and running underbalanced completions. A hydraulic snubbing unit is used to perform this operation.	Equipment failure (e.g. loss of pressure control, slip failure) leading to exposure to reservoir pressure and surface release of well fluids (potential toxic gas /fire/explosion hazard)	Personnel: 2-3 PIIIs or more based on application and reservoir characteristics	Note: Due to special hazards and rare use of this operation, a job specific risk assessment should be conducted to identify specific risk mitigation measures.
	Intervention			Property Damage: Potential asset damage due to loss of well control.	

CMSs Available for High Risk Activities Covered under W-7

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Robust Management of Well Servicing Operations (W-3) 1 CMS	Contractors Competency	A fully capable, trained crew for each type of operation. This is typically demonstrated by having, at a minimum, these programs: SSE program, Equipment operations & HSE training, Well & Job Type Knowledge,	Implemented and in compliance as applicable
	Oxy Well Site Managers	Onsite Supervision during high risk operational period (during the use of explosives, well or pump pressures exceeding 5000 psi, and establishing well control)*	
	HESMS PSPG's & BU Performance Standards	JSA's, BBS, PPE, Safe Work Systems and/or Work Permits, Hazard ID - HSE Exhibit Compliance with API RP 54.	
	Equipment	Equipment Acceptance Criteria (MIT/PM/ Inspection Programs in place), Operating equipment within OEM specifications and compliance with API RP 54, 8B, and 49, if H ₂ S is greater than 10 ppm as well as API Specs 4F, 8A and 8C.	
	Operating PSPG's	- 60.630.010 STD Well Control for Coiled Tubing, Snubbing, and Wireline Operations (Oxy & Contractor) - SimOps Procedures (Oxy & Contractor) - Weather (sustained winds above 35 mph or gust above 50 mph; blinding snow/rain/dust/sand; sleet; lightning; severe road or equipment icing, severely muddy roads; tornado/hurricane/flood warnings).*	Implemented and in compliance as applicable or approved OOG variance in effect

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Materials Handling, Rigging & Lifting, and Dropped Objects Controls (W-7) 1 CMS	Worksite Controls	- Well site induction for visitors - No-Go Zones designated in the lifting areas - Site traffic control and equipment staging plan (including use of trained spotters)	Implemented and in compliance as applicable
	SOP, JSA and Permits	- Written standard operating procedures (SOP) including for applicable specialized operations such as: - Fishing & Jarring - Sand Cleanout with Hoist - Perforating Gun Lifting - Contractors and Oxy personnel review and follow the applicable SOPs - Taglines in use for all lifting operations - Dropped Objects Management Program in place - JSA and TBT performed and necessary permits are obtained	
	Equipment	Lifting, rigging and material handling equipment meets the following: - Certified, tested and inspected per required frequency - Onsite management of lifting and rigging equipment via a controlled register - Equipment is rated for required load - Secondary retention and safety securing devices installed on applicable components at height	
	Personnel Competency	- Roles & responsibilities during lift operations, vehicular movement and staging clearly defined - Certificate of competence available for personnel involved with lifting activities, including crane and forklift operators	

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Control of Pumping / High Pressure Testing (W-8) 1 CMS	Worksite Controls	- Well site induction for visitors - Buffer Zone located around high pressure line of fire areas and non-essential personnel not admitted in the buffer zone	Implemented and in compliance as applicable
	SOP, JSA and Permits	- Contractors and Oxy personnel review and follow the applicable standard operating procedures (SOP) - JSA and TBT performed and necessary permits are obtained	
	Equipment	High Pressure Pumping/Testing equipment meets the following: - Certified equipment tested annually by 3rd party - Pumping iron and equipment to be rated for the maximum anticipated pressure - Pressure safety device in use and set for required pressure not to exceed the maximum pressure rating of pumping equipment - Iron & hose whip restraints appropriate for pressure rating secured at each connection	
	Contractor Competency	High pressure pumping / testing contractors/companies are audited at least every 2 years to ensure compliance with industry standards	

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Explosives Handling (W-9) 1 CMS	Worksite Controls	- Well site induction for visitors - No-Go Zones designated in the lifting areas - Non-use of radio and mobile phones on location until perforating guns are disarmed	Implemented and in compliance as applicable
	SOP, JSA and Permits	- Contractors and Oxy personnel review and follow the applicable standard operating procedures (SOP) - JSA and TBT performed and necessary permits are obtained	
	Equipment	- Compliance with API RP 67 for Oilfield Explosives Safety - MTTF calculations with set limit on number of runs allowed for all components - Transport skid used for mobilizing guns to well site	
	Contractor Competency	- Contractor crew handling explosives must have specialized training - Perforating Gun contractors/companies are audited at least every 2 years to ensure compliance with industry standards	

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Control of Flowback, Well Clean-up & Well Testing Operations (W-10) 1 CMS	Worksite Controls	- Well site induction for visitors - No-Go Zones designated around the flare and frac/open top tank - Buffer Zone located around high pressure line of fire areas and non-essential personnel not admitted in the buffer zone - Compliance with OOG spacing standard ES-10.400.502 for flare, pit and general equipment siting	Implemented and in compliance as applicable
	SOP, JSA and Permits	- Standard operating procedures (SOP) to include steps for each distinct operating phase - Contractors and Oxy personnel review and follow the applicable SOP - JSA and TBT performed and necessary permits are obtained (especially for hot work related to pipe make-up etc.)	
	Equipment	Flowback, Well Clean-up & Well Testing equipment meets the following: - Certified equipment tested annually by 3rd party - Applicable equipment rated for the maximum anticipated pressure and H₂S concentration (NACE) - Iron & hose whip restraints appropriate for pressure rating secured at each connection - Flare sited downwind of prevailing wind direction; flare equipped with remote auto-ignition capability & flare dispersion study completed for flammable/toxic effects - Frac Tanks, Open Top Tanks and tank trucks are grounded and bonded	
	Personnel Competency	Flowback, Well Clean-up & Well Testing contractors/companies are audited at least every 2 years to ensure compliance with industry standards	

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Radioactive Source Controls (W-11) 1 CMS	Worksite Controls	- Well site induction for visitors - No-Go Zone designated in the radioactive source loading are	Implemented and in compliance as applicable
	SOP, JSA and Permits	- Contractors and Oxy personnel review and follow the applicable standard operating procedures (SOP) - JSA and TBT performed and necessary permits are obtained - Compliance with regulatory requirements and industry safety standards (e.g., IAEA RS-G-1.1) for logging with radioactive sources	
	Equipment	- Compliance with regulatory requirements and industry safety standards (e.g., IAEA RS-G-1.1) for logging with radioactive sources	
	Contractor Competency	- Contractor crew handling radioactive source must have specialized training - Radioactive source contractors/companies are audited at least every 2 years to ensure compliance with regulatory standards	

Event Scenario 8-1: Well Servicing Mobilization and Demobilization (Land-Based Rigs, 2 CMSs Available)

This Event Scenario will not require risk ranking (considered C Level or less) if no more than 6 vehicles/self-propelled equipment (only vehicles larger than pickup size vehicles are counted) on or near the location at one time.

- **Consequence Level: typically Level 3 (1 PII) or 4 (2-3 PIIs) based on local conditions.**

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Robust Rig Movement Planning and Supervision (W-4) 1 CMS	Written Well Servicing Mobilization Plan	Equipment mobilization will not occur if hazards increase due to: - Weather (e.g., sustained winds above 35 mph or gust above 50 mph, blinding snow/rain/dust/sand, sleet, lightning, severe road or equipment icing, severely muddy roads, tornado/hurricane/flood warnings) - Terrain (inadequate road/route preparation or road condition) - Time limits: If distance is too far to complete in a maximum 14 hour work day, prepare & plan overnight staging areas	Implemented and in compliance as applicable or approved OOG variance in effect
		Moves are sequenced to minimize site congestion and multiple lifts.	
		Site schematics with equipment layout are designed and utilized by contractors	
		Cranes are certified for equipment weights and have qualified operators & riggers. * Lifting devices have current certification & appropriate for lift.	
		Designated spotters with appropriate training and communications capabilities to truck drivers, equipment operators & other directly involved parties.	
		Routes on Oxy property are driven/walked to identify potential hazards.	
	Controls	Contractor project supervisor on location monitoring equipment mobilization	Implemented and in compliance as applicable

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Rig Design (W-5) 1 CMS	Modular Rigs	Modular Rigs (human intervention to move and rig up is reduced) Equipment requires 3 or less staff during rig up operations Pick loads are minimized through bundling equipment (e.g. pump and tank on same skid, carrier mounted, wheeled (does not require trailer to move) Equipment tool skids that reduce the number of picks Human hoisting activities are eliminated Equipment is an integral part of the rig eliminating need for hoisting	Implemented and in compliance as applicable

Event Scenario 8-2: Well Servicing Mobilization and Demobilization (Offshore Rigs, 3 CMSs Available)

Mobilization-De-mobilization only includes the barge movement & pre-load, not RU/RD which is covered under the Higher Risk Activities Register. Barge Move refers to mobilization and demobilization collectively.

- **Consequence Level typically Level 4 (2-3 PIs) or 5 (>3 PIs) based on ship collision impact to platform leading to loss of containment (possible fire, explosion, toxic event); PD-LOR Consequence Level 4 or 5 based on location**

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
Planning & Personnel (W-12) 1 CMS	Barge Rig Move Procedure	Location approval by Barge Marine Superintendent (position, water depth, expected penetration (historical data footprints, soil sample for new location), stand-off distance)	Implemented and in compliance as applicable
		Weather forecast (2 independent sources)	
		Seabed survey – cleared area defined – pipelines marked where necessary – debris cleared	
		Weather criteria for move – barge move not allowed if wind speed above 15 knots or sea state 5 feet or higher	
	Barge Move Planning Meeting & Pre-Move Checks	Barge Rig Crew undertake pre-move checks as per barge rig move procedure	
		Watertight Integrity Confirmed - Deck cargo secured - readiness to move confirmed by Master & Chief Engineer	
		Variable deck load calculated and within operating parameters	
		Barge rig move briefing with senior personnel on barge	
		Contractor Emergency Response Plan approved by Oxy	
	Minimum Qualified Personnel	Licensed Vessel Crews Rig minimum manning level as per Flag State Requirements (e.g. ABS) Oxy Rep Onboard Licensed OIM Onboard Barge Crew with Emergency Response Training	
Barge Move Controls (W-13) 1 CMS	Procedural & Functional Controls	Move-on/Move-off undertaken in daylight hours only	Implemented and in compliance as applicable
		Barge confirmed in final acceptable position	
		Barge commences preload operation at 5 foot air gap	
		During preload operations - no helicopter & crane operations allowed	

CMS Name	Mitigation Element	Standards and Requirements	Mitigation Compliance
		No personnel or vessels allowed alongside jacket until preload is dumped	
		ESD function tested and handed to Barge. ESD manned during jacking operations	
		PM and 3rd party inspection on jacking components - redundancy built into jacking system	
Barge Propulsion & Support Equipment (W-14) 1 CMS	Certified equipment	Self-propelled barge and approaching jacket at < 1 Knot	Implemented and in compliance as applicable
		Redundant generator in operation during Barge move	
		Emergency response & evacuation equipment available	

Appendix F – Business Unit Annual Risk Inventory Report Template

EXECUTIVE SUMMARY²

Risk Inventory Summary

	<i>(last year)</i>			<i>(current year)</i>			<i>(next year)</i>		
		Validated	Residual		Validated	Residual		Validated	Residual
C with PEC									
D Level Risks									
E Level Risks									
Mitigation Spending/Budget for these Specific Risks*	\$			\$			\$		

* In order of preference use actual amounts, good faith estimate or indicate “not available”.

Risk Mitigation Spending and Budgets by Facility

BU Risk Project Budgets by Facility (in thousands) for <i>(next year)</i> (mandatory for C w/PEC, D and E level risks)							
	Siting/ Layout	Fire Protection	ESD/ Isolation	Relief/ Flare	<i>(as needed)</i>	Misc.	Totals
<u>(Facility 1)</u>	\$	\$	\$	\$	\$	\$	\$
<u>(Facility 2)</u>	\$	\$	\$	\$	\$	\$	\$
<u>(Facility 3)</u>	\$	\$	\$	\$	\$	\$	\$
<u>General Facilities</u>	\$	\$	\$	\$	\$	\$	\$

Regulatory Programs and Other PECs

OSHA PSM Facilities (or State Equivalents)	
EPA RMP Facilities (or State Equivalents)	
DOT/PHMSA Regulated (and State Equivalents)	
Infrequent recurring conditions or circumstances in which PEC could arise (don't count those above)	

² This Risk Inventory Report summarizes the BU's reporting year and coming year's plans regarding higher-level HSE risks. In addition to reporting on D and E level risks and C level risks with PEC, BUs have been asked to estimate spending designated specifically for risk mitigation projects with respect to these risks. These spending estimates do not reflect all expenditures for risk mitigation in operations, functional programs, and imbedded in capital projects, but rather those designated for discrete mitigation activities to address specific risk scenarios.

1. SUMMARY OF MAJOR RISK MITIGATION ACCOMPLISHMENTS IN (current year)

(Using a maximum of ½ page, describe highlights of what has been accomplished to reduce the BU's risk profile)

2. SUMMARY OF MAJOR RISK MITIGATION PLANS FOR (coming year)

(Using a maximum of ½ page, describe highlights of what will be accomplished in the coming year to reduce the BU's risk profile)

3. POTENTIAL ELEVATED CONSEQUENCES

(Summarize any C w/PEC, D and E level risks that have PECs. Also, provide a qualitative summary of infrequent but recurring conditions or circumstances in which PEC could arise, such as atypical Facility activities, meteorological conditions or community events near a Facility.)

RISK PROGRAM STATUS

(The objectives of the remainder of the annual report are for BU's management to understand their HSE risks and to better manage their risk mitigation strategy. This will also be used by OOG HSE to better understand and communicate the BU's risks to executive management. The information should be presented in a concise manner at an appropriate level for the BU GM to understand the issues and develop their risk mitigation strategy.)

1. OVERVIEW OF BU'S OPERATIONS

(Brief overview of the BU's operations – type and number of facilities such as production areas, offshore platforms, gas plants, etc.)

2. SUCCESSES AND CHALLENGES

(Highlight major activities and program changes in the current year that will improve effectiveness and efficiency. If applicable, provide a short review any major obstacles to meeting risk program objectives and how they are / will be addressed.)

3. MAJOR INITIATIVES PLANNED FOR 201Y

(Highlight major activities and program enhancements planned for the coming year.)

4. REGULATORY RISK PROGRAM STATUS

(Highlight the risk regulations governing the facility, process or operations and provide a BU regulatory status.)

<i>Facility</i>	<i>Regulatory Risk Program</i>	<i>Basis for Applicability</i>	<i>Scenarios addressed (RMP Programs only)</i>	<i>Recent or Planned Changes</i>

5. OPEN RISK REGISTERS

(Provide a list of risk registers that remain open at the end of the reporting year.)

Risk ID	Risk Title	Initial Risk Level	Current Risk Level	Target Risk Level	Mitigation Phase

6. CLOSED RISK REGISTERS

(Provide a list of risk registers that were closed during the reporting year.)

Risk ID	Risk Title	Initial Risk Level	Current Risk Level

7. ATTACHMENTS

Oxy Risk Detail (One-page) Report for each C w/PEC, D and E level risk that are not Residual Risks must be attached. See example below.

Oxy Risk Detail (insert BU name – facility name)

Original Risk Level:

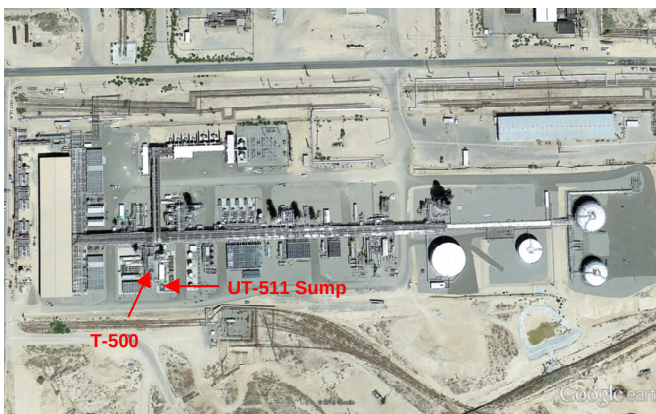
D - Onsite Personnel

Risk Description:

Inadvertent opening or valve leak-through of 2" valve on bottom of T-500 causes amine flow and gas blow-by into sump UT-511.

Potential Consequences:

- Injury to persons in occupied buildings due to blast over-pressure damage or to persons in proximity to blast / fire



Current Risk Level:

D - Onsite Personnel

Existing Mitigations:

- Prevent release: high level alarm with operator response
- Prevent initiation: existing 2" drain valve is car-sealed closed
- Prevent ignition: proper electrical classification



Target Risk Level:

C - Onsite Personnel

Risk Mitigation Action Plan:

- Add second car-sealed 2" drain valve and add isolation valve on overhead - next turnaround.
- Add valve and/or a spectacle blind on the overhead line coming out of the T-500 vessel in order to allow for isolation and depressurization of the vessel, whenever the vessel is not in service. (\$150K, 3/2014)

OOG Risk Matrix						
Probability	5	B	C	D	E	E
	4	B	C	C/D	D	E
	3	A	B	C	D	D
	2	A	B	B	C	D
	1	A	A	B	B	C
		1	2	3	4	5
Consequence						

Responsible Managers

- Operations Manager
- Plant Manager
- HSE Manager

Risk Engineering

- Risk Engineer